

機械手臂 順向運動學

Manipulator Forward Kinematics

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引言 -1

□ 運動學(Kinematics)：討論運動狀態本身，未連結到產生運動的「力」

◆ 位置(x)、速度(v)、加速度(a)、和時間(t) 之間的關係

◆ 移動/轉動

◦ 位置/姿態

◦ 速度/角速度

◦ 加速度/角加速度

$$\begin{aligned} v &= \frac{d}{dt}x & a &= \frac{d}{dt}v \\ a &= \frac{d^2}{dt^2}x & vdv &= ads \end{aligned}$$

□ 動力學(Dynamics)：討論力/力矩如何產生運動

◆ Newton's 2nd Law

$$\sum F = ma$$

◆ Work & energy

$$T_1 + V_1 + U_{1-2}' = T_2 + V_2$$

◆ Impulse & momentum

$$\int \sum F dt = G_2 - G_1$$

引言 -2

□ 機械手臂

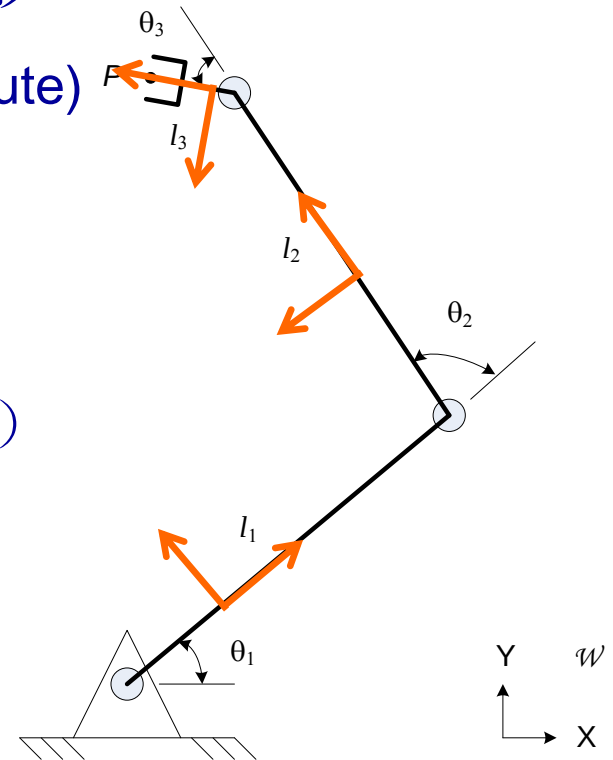
- ◆ 多個桿件(link)相串連，具有複雜的幾何外形
- ◆ 桿件間可相對移動(prismatic)或轉動(revolute)
由致動器驅動來達成

□ 對應關係

- ◆ 需求：手臂末端點狀態（位置 ${}^W P$ 、速度...）
 - ◆ 達成方式：驅動各致動器
- $${}^W P = f(\theta_1, \theta_2, \dots, \theta_n)$$

□ 描述手臂狀態方法

- ◆ 找出桿件間的相對幾何狀態
- ◆ 在各桿件上建立frame，以frame狀態來代表桿件狀態



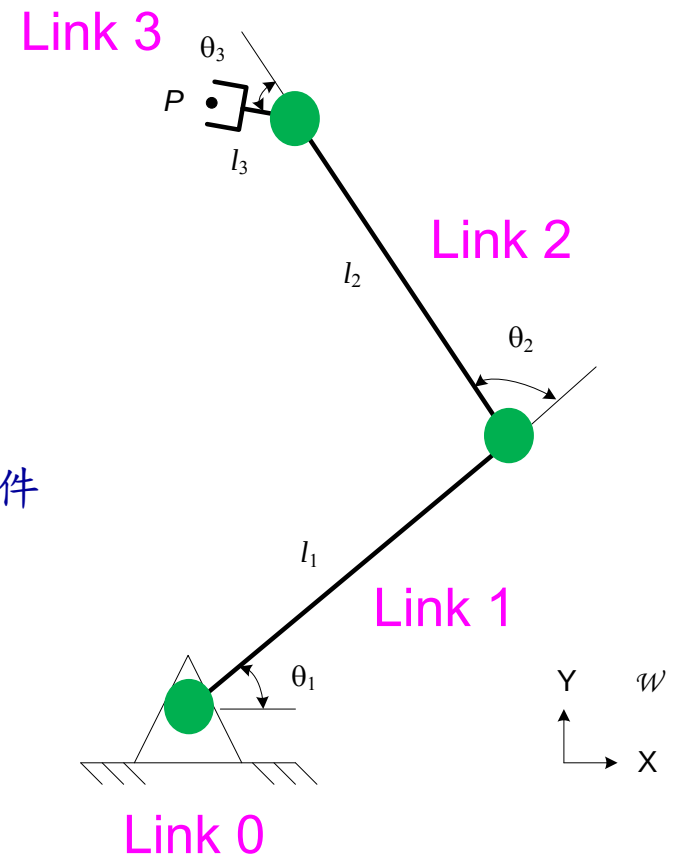
手臂幾何描述方式 -1

□ Joint

- ◆ 每個revolute或prismatic的joint具有 1 DOF
- ◆ 每個joint對 某特定axis 進行rotation或translation

□ Link

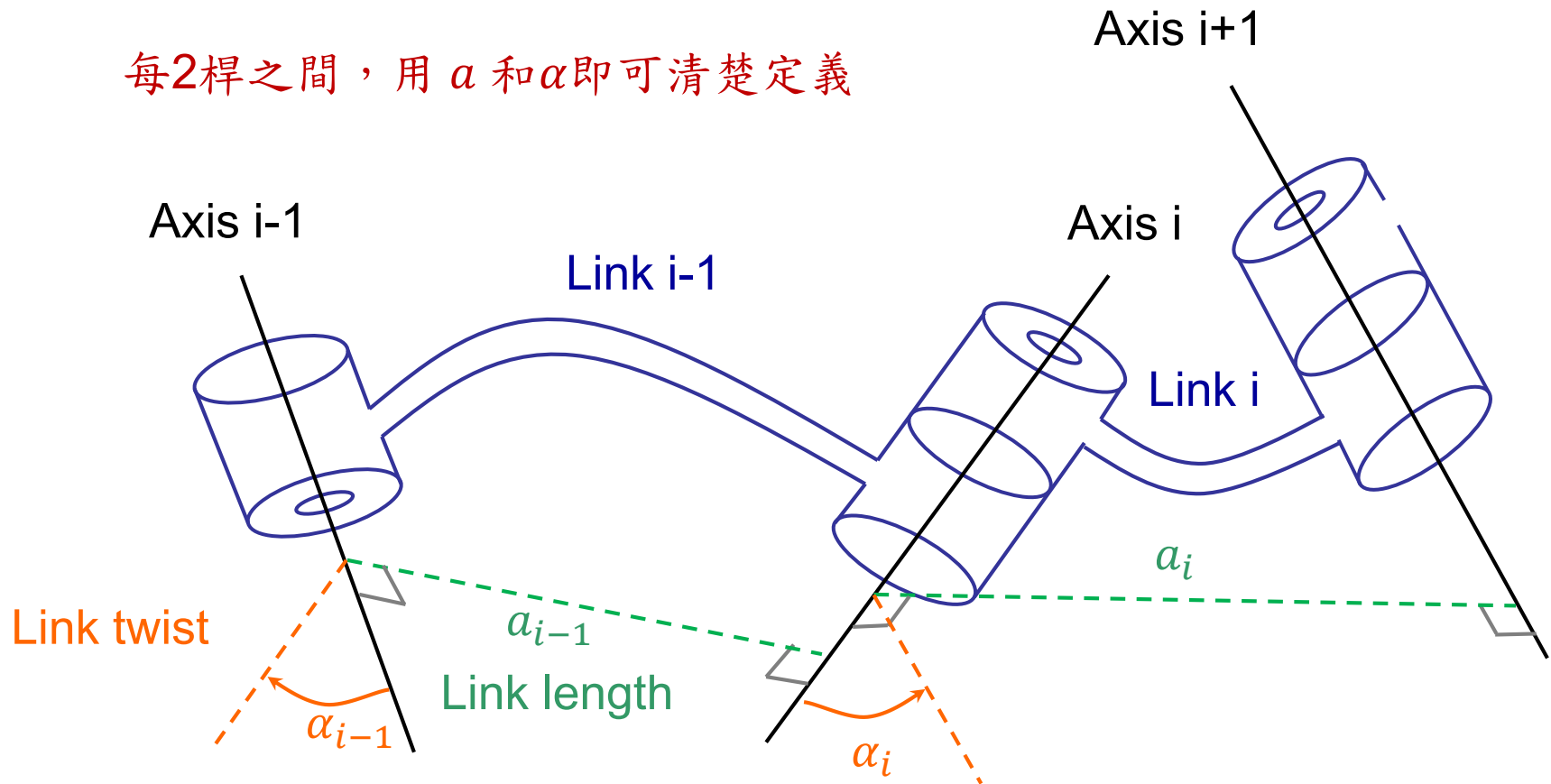
- ◆ 連接joints的桿件，為剛體(rigid body)
- ◆ 編號方式
 - Link 0: 地桿，不動的桿件
 - Link 1: 和Link 0相連，第一個可動的桿件
 - Link 2: 第二個可動的桿件
 - 依序下去...



手臂幾何描述方式 -2

- 對空間中2個任意方向的axes，兩axes之間具有一線段和此2個axes都相互垂直

每2桿之間，用 a 和 α 即可清楚定義



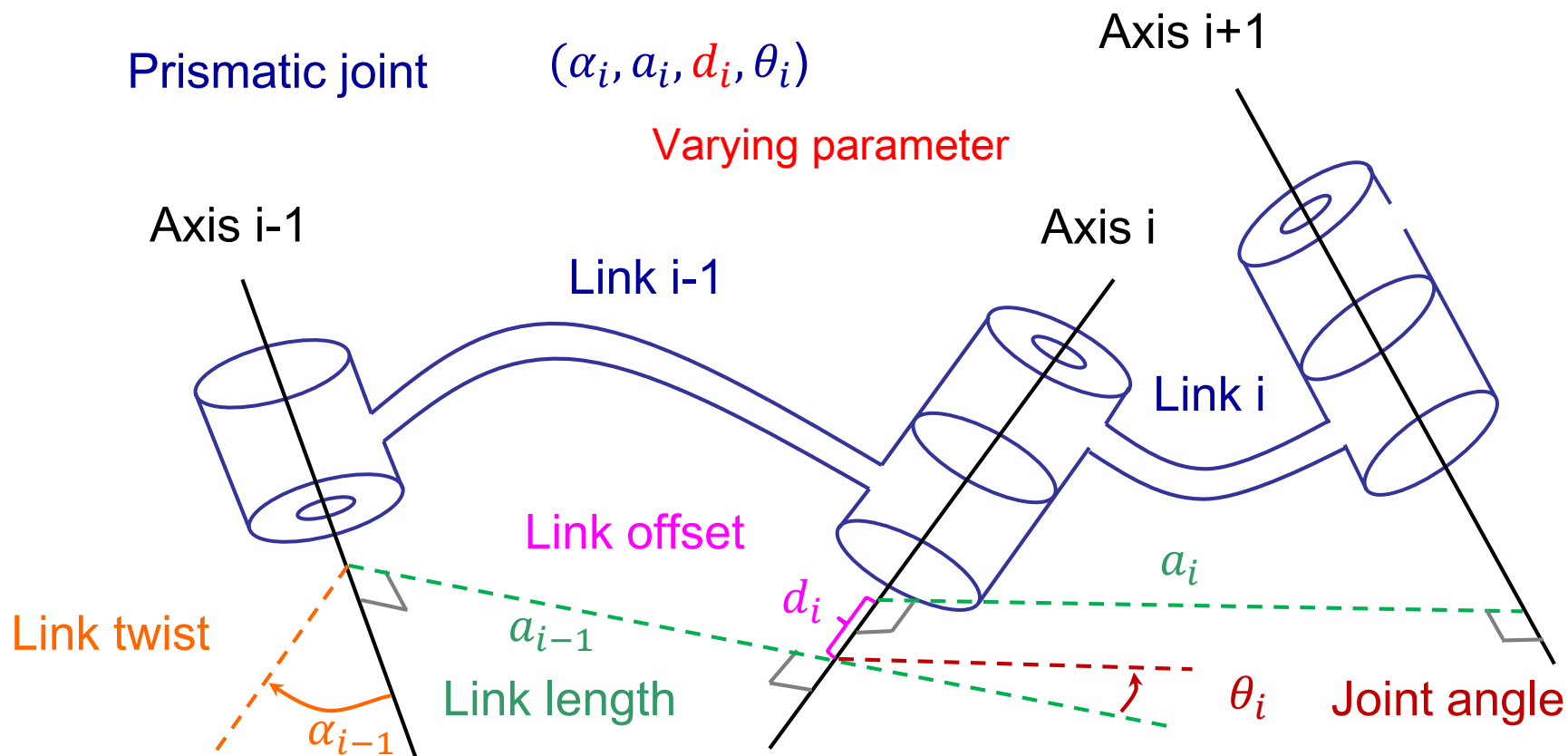
手臂幾何描述方式 -3

- 但若要多桿串連，則另需要兩個參數(d_i, θ_i)，來描述相鄰線段 a_{i-1} 和 a_i 間的相對幾何關係

Revolute joint $(\alpha_i, a_i, d_i, \theta_i)$

Prismatic joint $(\alpha_i, a_i, d_i, \theta_i)$

Varying parameter



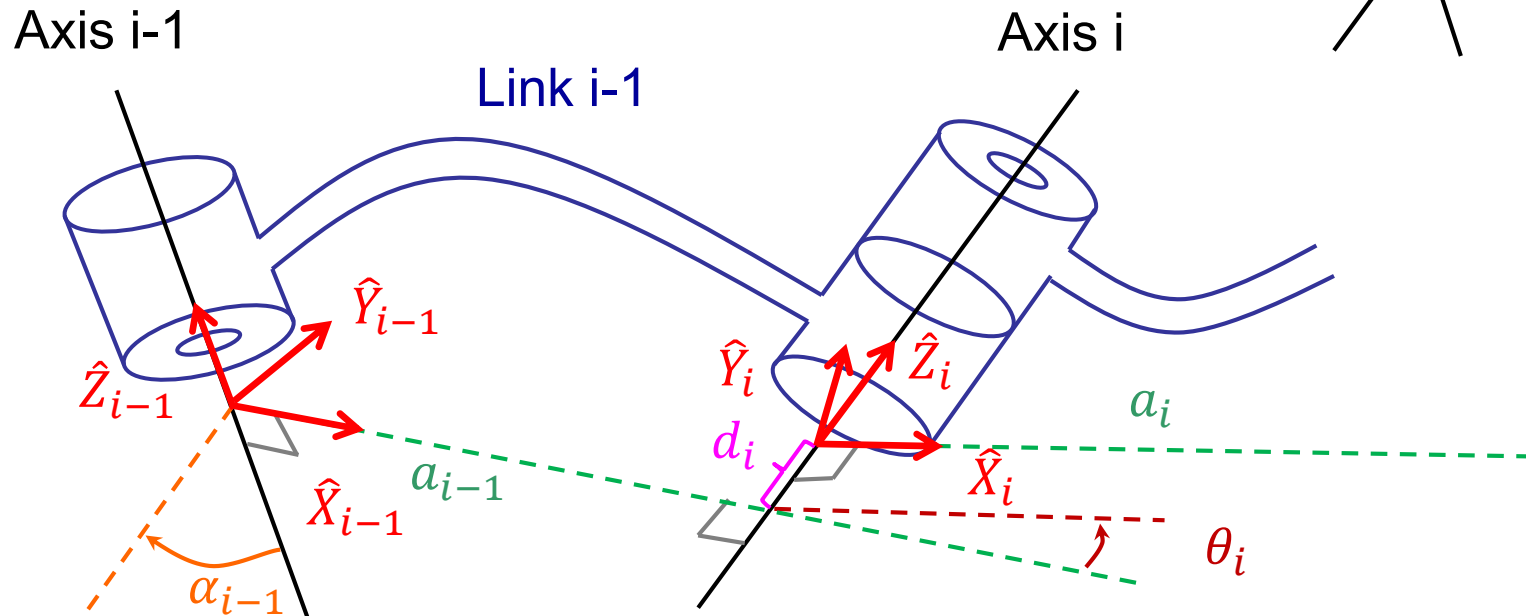
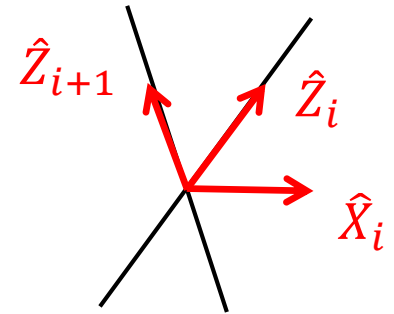
桿件上建立Frames -1

□ $\hat{Z}_i$ 轉動或移動axis的方向

□ $\hat{X}_i$ 沿著 a_i 方向 (if $a_i \neq 0$)

和 $\hat{Z}_i$ 和 $\hat{Z}_{i+1}$ 兩者垂直 (if $a_i = 0$)

□ $\hat{Y}_i$ 與 $\hat{X}_i$ 和 $\hat{Z}_i$ 兩者垂直，依循右手定則



桿件上建立Frames -2

□ 地桿 link (0)

Frame {0} coincides with frame {1}

$$a_0 = 0$$

$$\alpha_0 = 0$$

Revolute joint

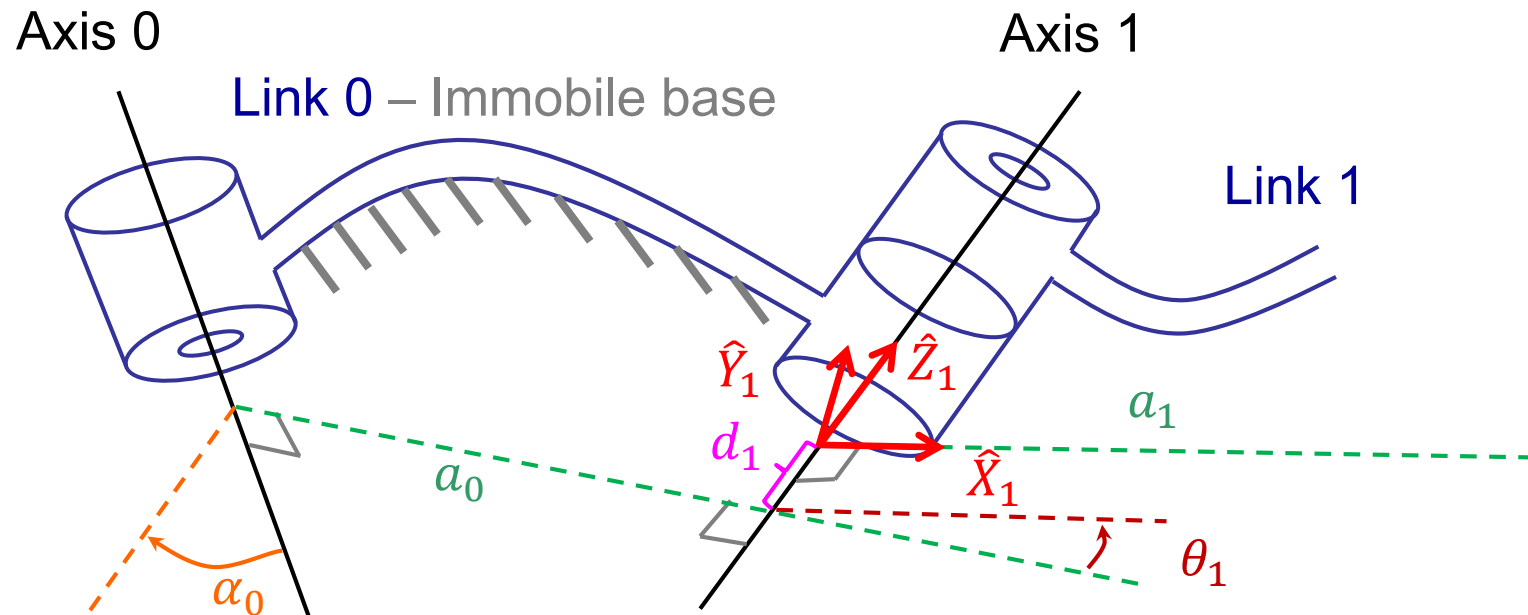
θ_1 arbitrary

$$d_1 = 0$$

Prismatic joint

d_1 arbitrary

$$\theta_1 = 0$$



桿件上建立Frames -3

□ Last link (n)

取和 $\hat{X}_{n-1}$ 同方向

$$a_n = 0 \quad \alpha_n = 0$$

Revolute joint

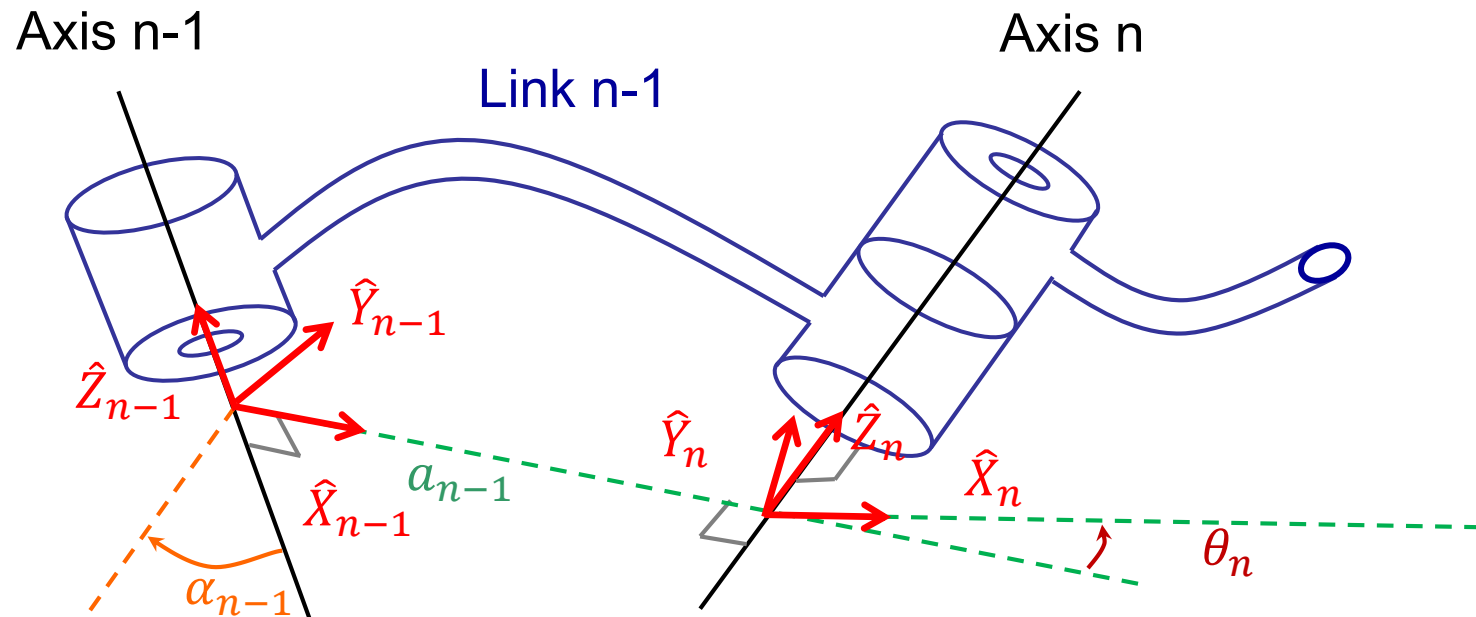
θ_n variable

$$d_n = 0$$

Prismatic joint

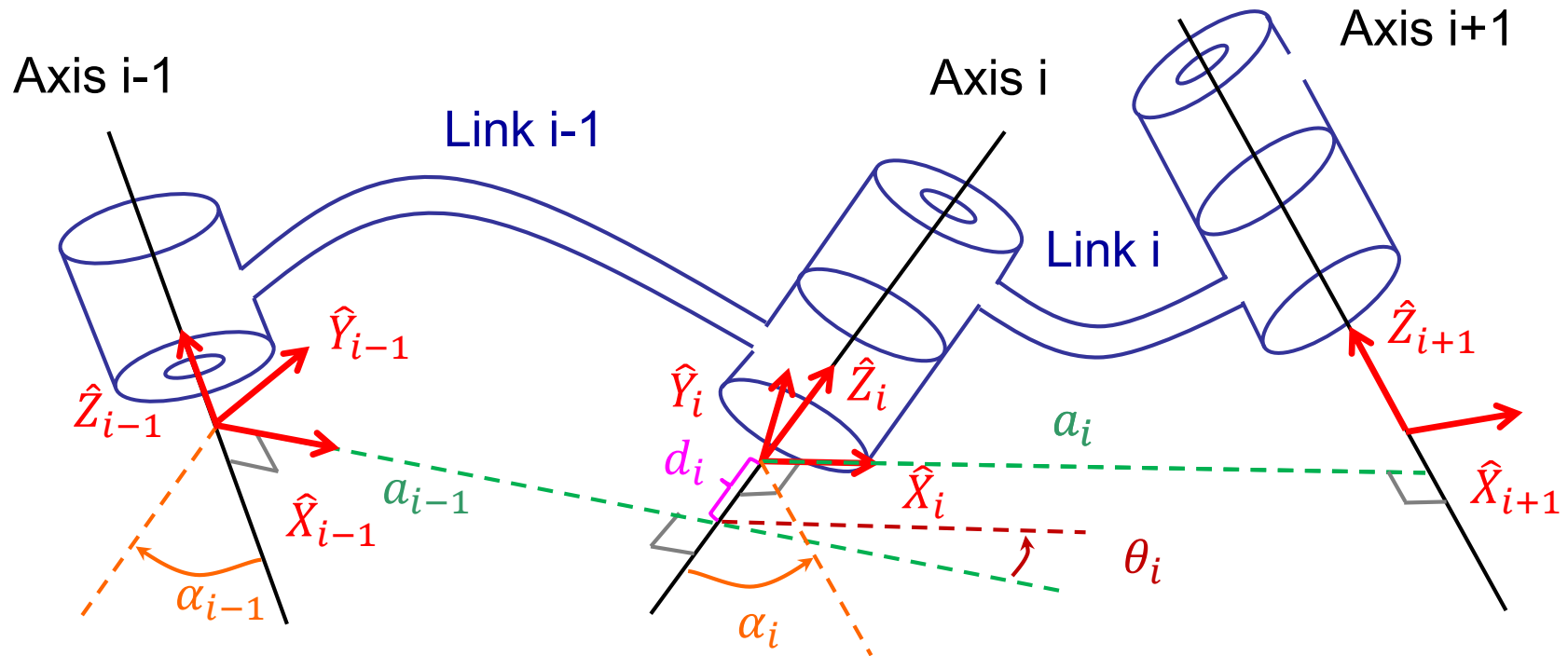
d_n variable

$$\theta_n = 0$$



Denavit-Hartenberg表達法 (Craig version)

- α_{i-1} : 以 $\hat{X}_{i-1}$ 方向看, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的夾角 α : 连杆扭角, 轴线之间的夹角
- a_{i-1} : 沿著 $\hat{X}_{i-1}$ 方向, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的距離 ($a_i > 0$) a : 连杆长度, 垂线长度
- θ_i : 以 $\hat{Z}_i$ 方向看, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的夾角 θ : 关节角, x与垂线的夹角
- d_i : 沿著 $\hat{Z}_i$ 方向, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的距離 d : 连杆偏距, 垂线与基点之间的距离



Link Transformations -1

$${}^{i-1}P = {}^{i-1}_iT^iP$$

$${}^{i-1}P = {}^{i-1}_RT^R_QT^Q_PT^P_iT^iP$$

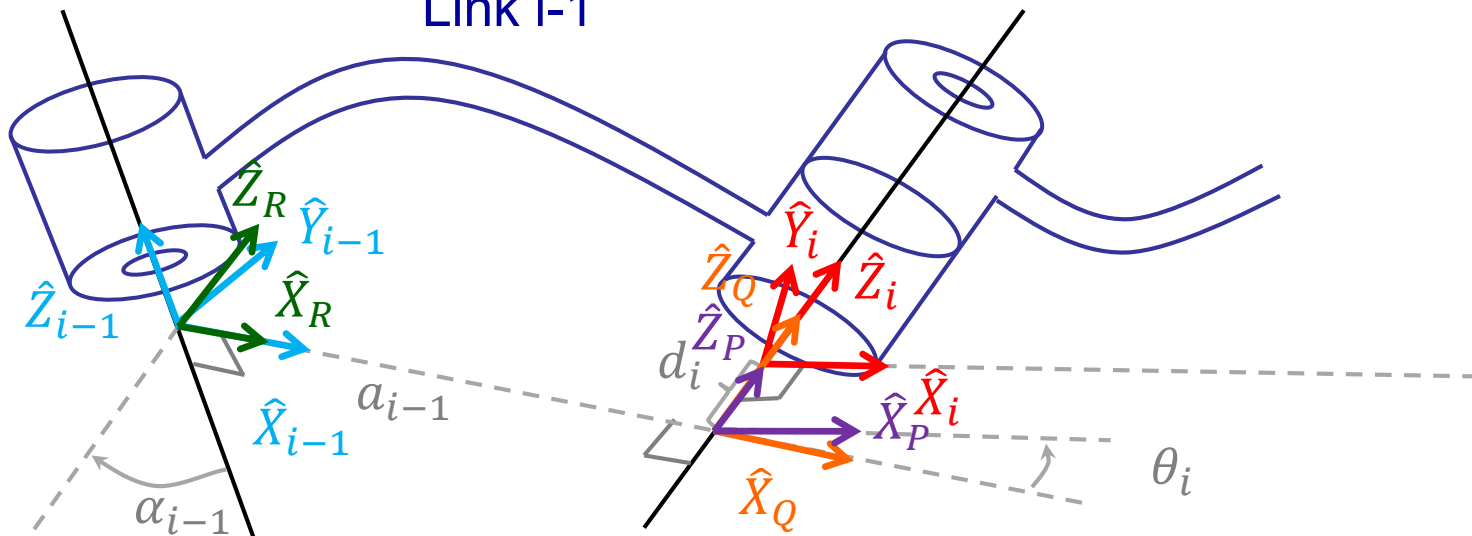
$$\begin{aligned} {}^{i-1}_iT &= {}^{i-1}_RT^R_QT^Q_PT^P_iT \\ &= T_{\hat{X}_{i-1}}(\alpha_{i-1})T_{\hat{X}_R}(a_{i-1})T_{\hat{Z}_Q}(\theta_i)T_{\hat{Z}_P}(d_i) \end{aligned}$$

1. $T_{\hat{X}_{i-1}}(\alpha_{i-1})$: 繞 $\hat{X}_{i-1}$ 軸旋轉 α_{i-1} (連桿扭角, Link twist)。
2. $T_{\hat{X}_R}(a_{i-1})$: 沿 $\hat{X}_R$ 軸平移 a_{i-1} (連桿長度, Link length)。
3. $T_{\hat{Z}_Q}(\theta_i)$: 繞 $\hat{Z}_Q$ 軸旋轉 θ_i (關節角, Joint angle)。
4. $T_{\hat{Z}_P}(d_i)$: 沿 $\hat{Z}_P$ 軸平移 d_i (連桿偏距, Link offset)。

Axis i-1

Link i-1

Axis i



Link Transformations -2

□ Thus

$${}^{i-1}_iT = T_{\hat{x}_{i-1}}(\alpha_{i-1})T_{\hat{x}_R}(a_{i-1})T_{\hat{z}_Q}(\theta_i)T_{\hat{z}_P}(d_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

□ 連續link transformations

$${}^0_nT = {}^0_1T {}^1_2T {}^2_3T \dots {}^{n-2}_{n-1}T {}^{n-1}_nT$$

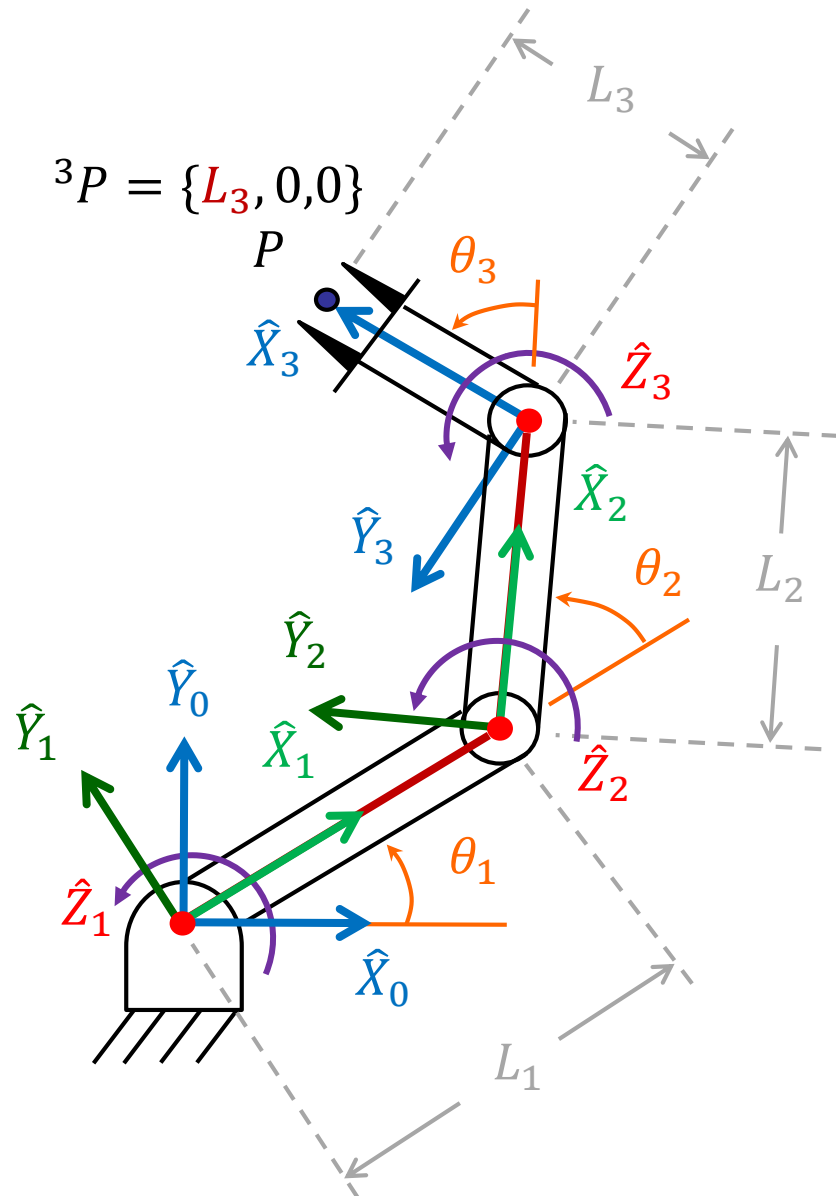
Frame {n} 相對於 Frame {0} 的空間幾何關係具清楚且量化之定義

在Frame {n} 下表達的向量可轉回 Frame {0} 下來表達

Example: A RRR Manipulator

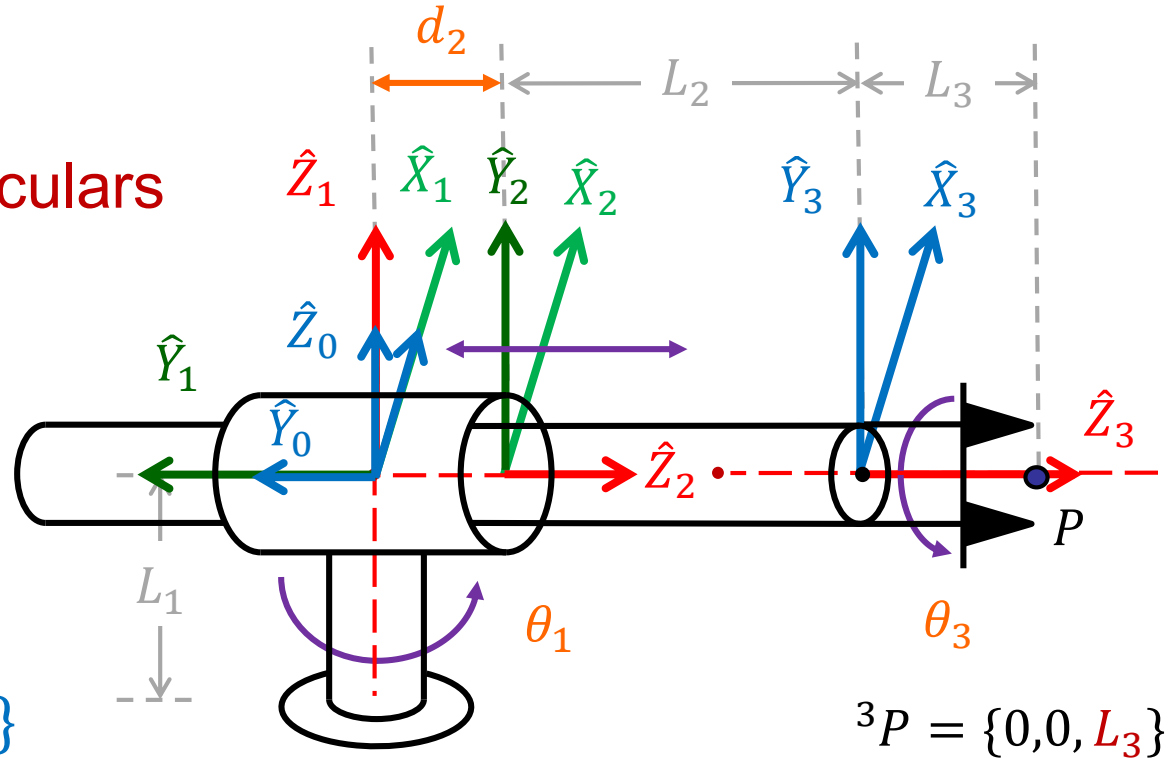
- Joint axes
- Common perpendiculars
- $\hat{Z}_i$
- $\hat{X}_i$
- $\hat{Y}_i$
- Frames $\{0\}$ and $\{n\}$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



Example: A RPR Manipulator -2

- Joint axes
- Common perpendiculars
- $\hat{Z}_i$
- $\hat{X}_i$
- $\hat{Y}_i$
- Frames $\{0\}$ and $\{n\}$



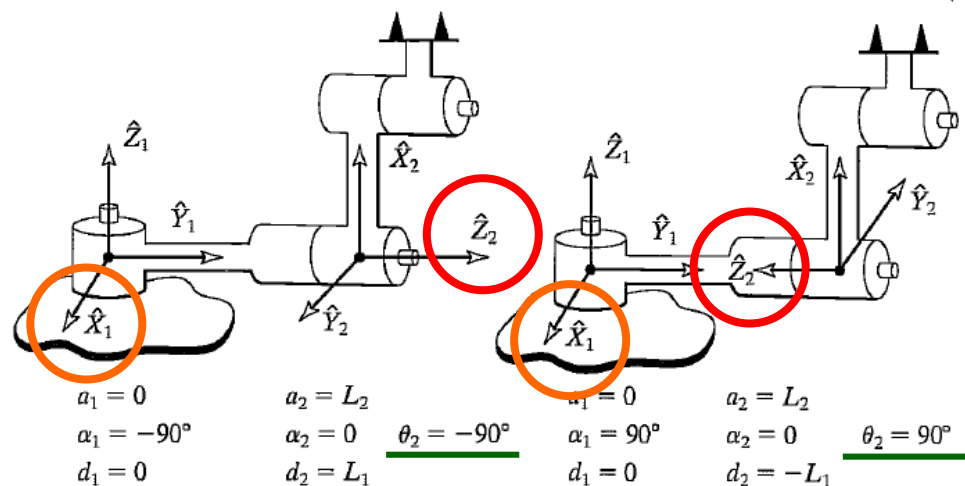
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90°	0	d_2	0
3	0	0	L_2	θ_3

Example: A RRR Manipulator -3

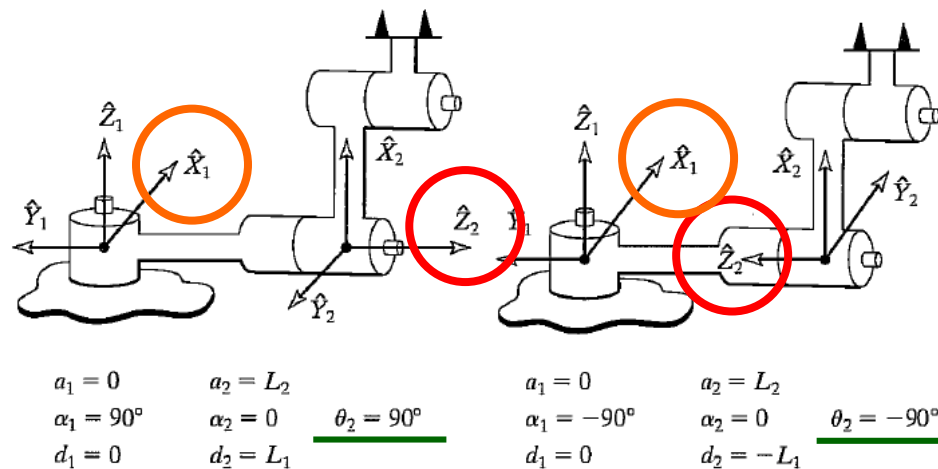
□ 當 $a_1 = 0$

$\hat{Z}_1$ 和 $\hat{Z}_2$ 相交

◆ $\hat{Z}_2$ 兩個選擇

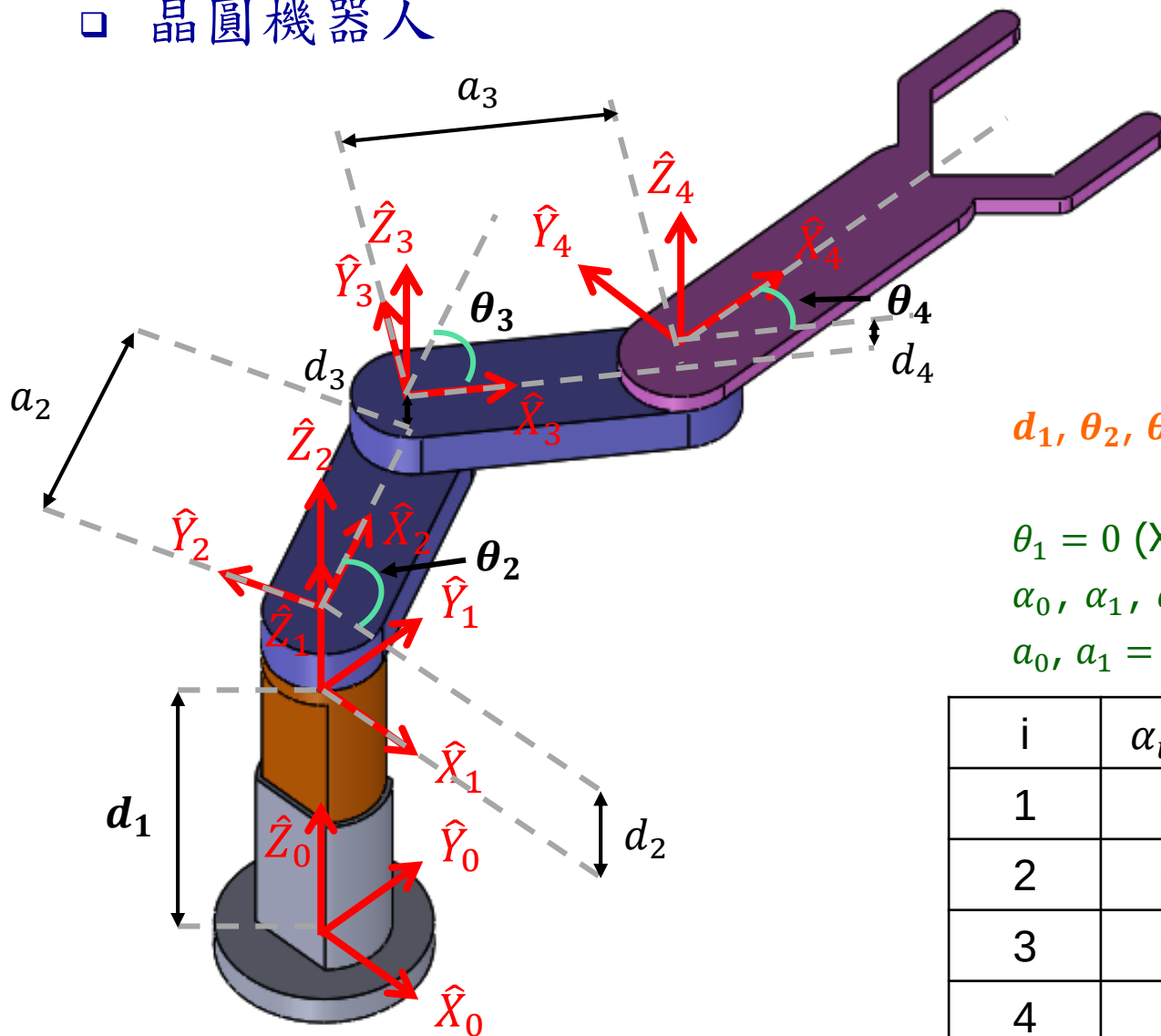


◆ $\hat{X}_1$ 兩個選擇



Example: A PRRR Manipulator

晶圓機器人



$d_1, \theta_2, \theta_3, \theta_4$ 為驅動關節

$\theta_1 = 0$ ($X_0 // X_1$)

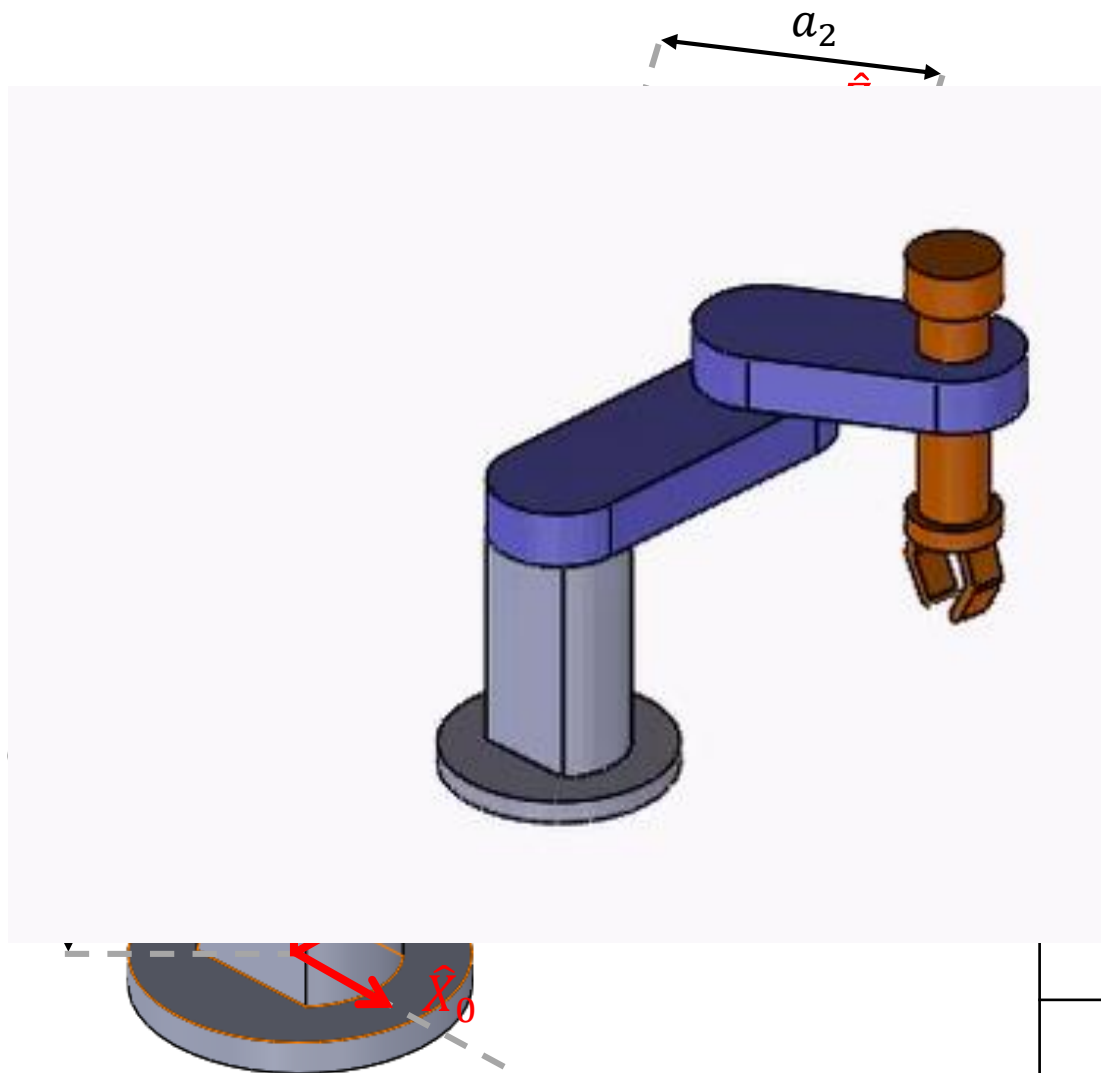
$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z軸相互平行)

$a_0, a_1 = 0$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	0	d_2	θ_2
3	0	a_2	d_3	θ_3
4	0	a_3	d_4	θ_4

Example: A RRRP Manipulator

SCARA 機器人



$\theta_1, \theta_2, \theta_3, d_4$ 為驅動關節

$\theta_4 = 0$ ($X_0 \parallel X_1$)

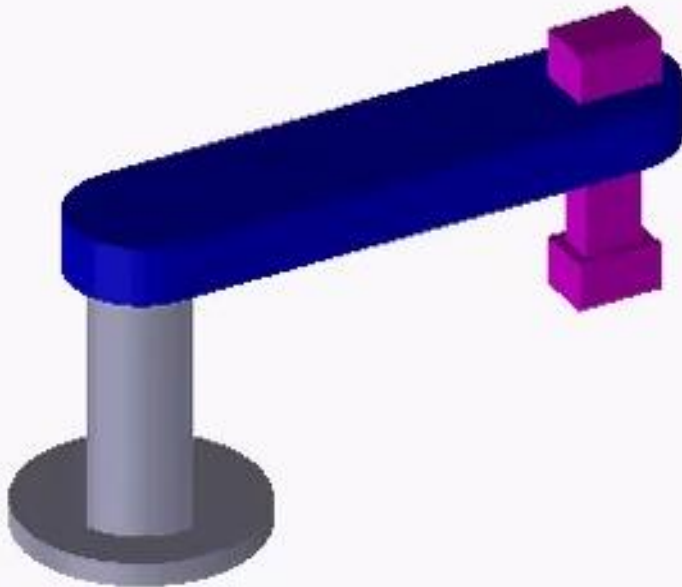
$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ (所有Z軸相互平行)

$a_0, a_3 = 0$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	θ_1
2	0	a_1	d_2	θ_2
3	0	a_2	0	θ_3
4	0	0	d_4	0

Example: A RP manipulator

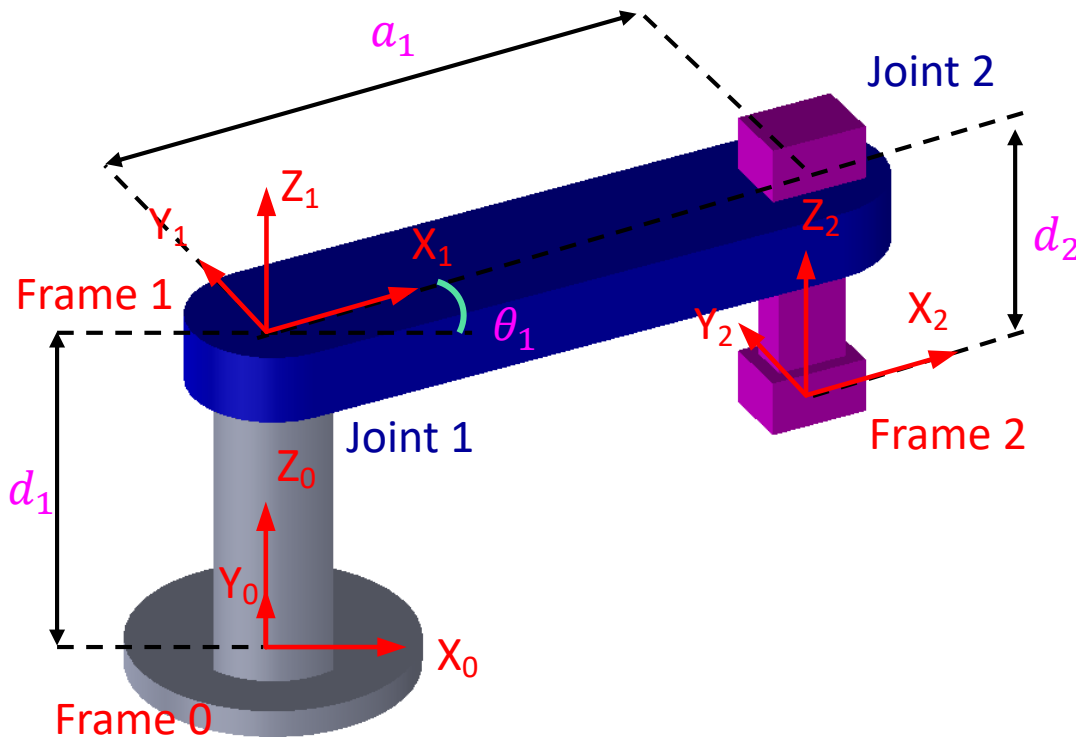
- In-video Quiz：下方手臂由一個revolute joint和一個prismatic joint組成，在所有的DH參數 $(a_{i-1} \alpha_{i-1} d_i \theta_i)$ 中，哪兩個參數為驅動關節？



- (A) α_1, a_2
- (B) α_1, d_2
- (C) θ_1, a_2
- (D) θ_1, d_2

Example: A RP manipulator (Solution)

- In-video Quiz：下方手臂由一個revolute joint和一個prismatic joint組成，在所有的DH參數 (a_{i-1} α_{i-1} d_i θ_i) 中，哪兩個參數為驅動關節？



For Joint 1:

$a_0 = 0$ (Z_0 Z_1 are concurrent)

$\alpha_0 = 0$ ($Z_0 \parallel Z_1$)

For Joint 2:

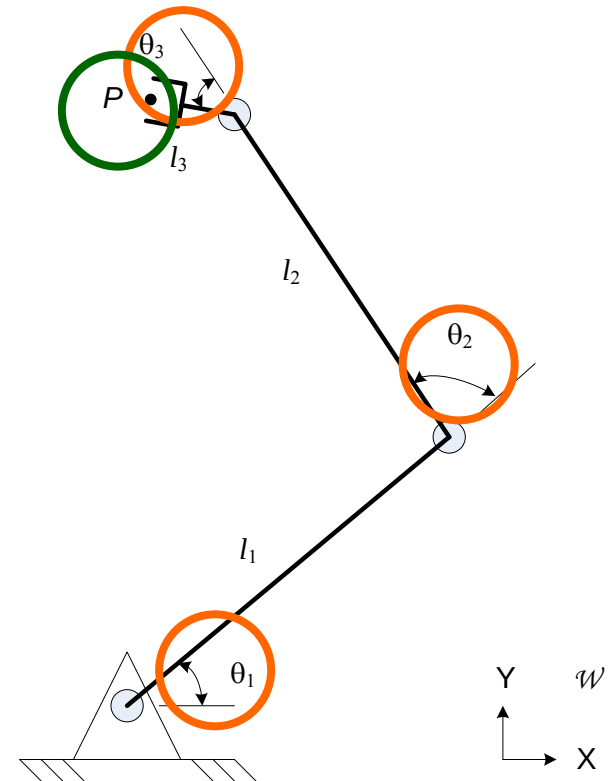
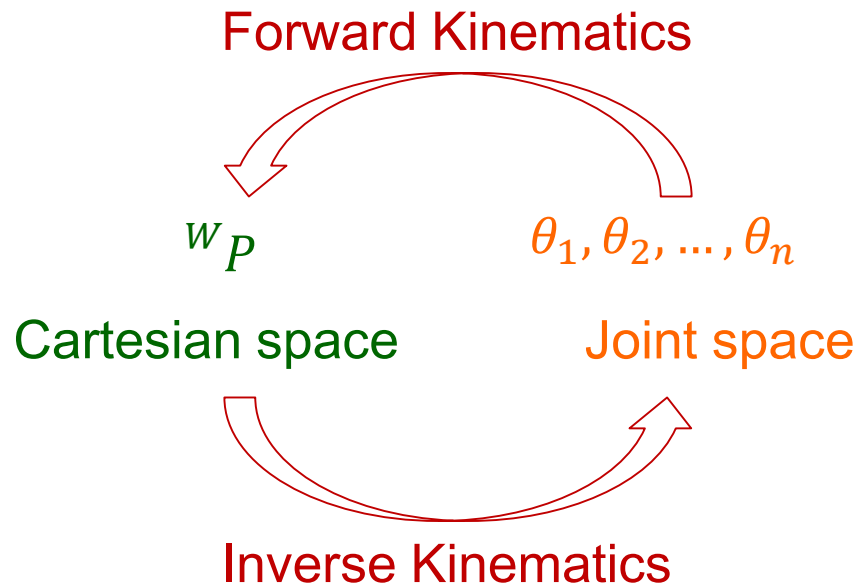
$\alpha_1 = 0$ ($Z_1 \parallel Z_2$)

$\theta_2 = 0$ ($X_1 \parallel X_2$)

θ_1, d_2 are varying parameters

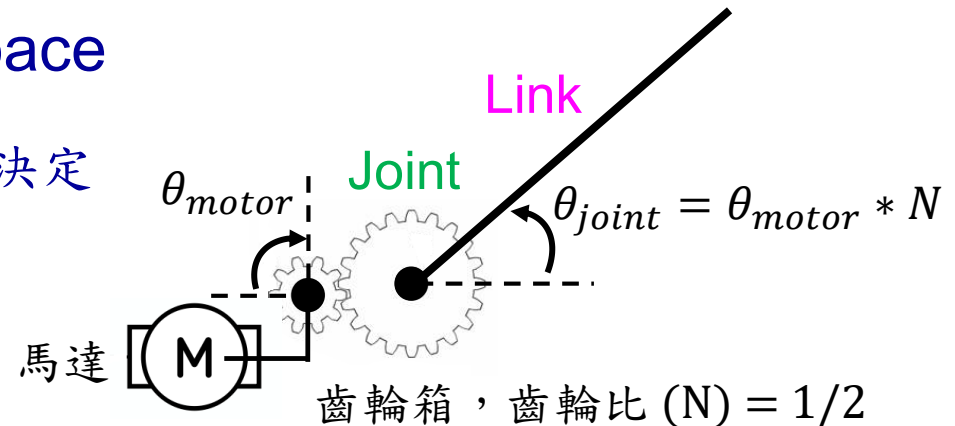
Actuator, Joint, and Cartesian Spaces -1

- Joint space $\Leftrightarrow$ Cartesian space



- Actuator space $\Leftrightarrow$ joint space

- 由連結致動器和joint的機構決定



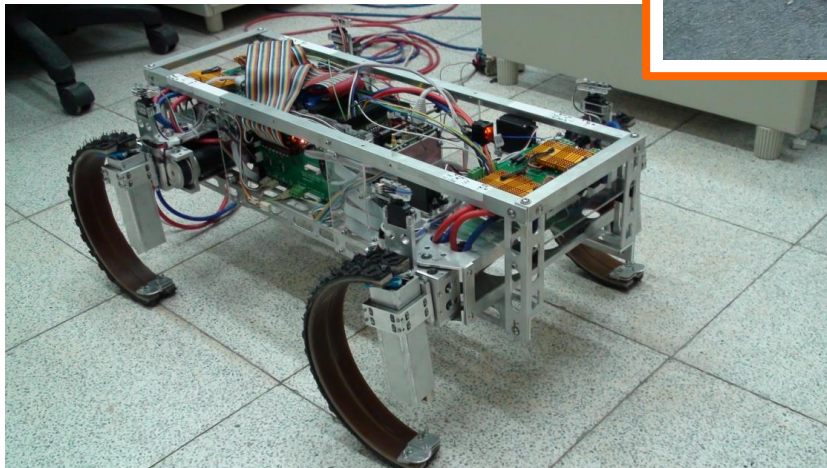
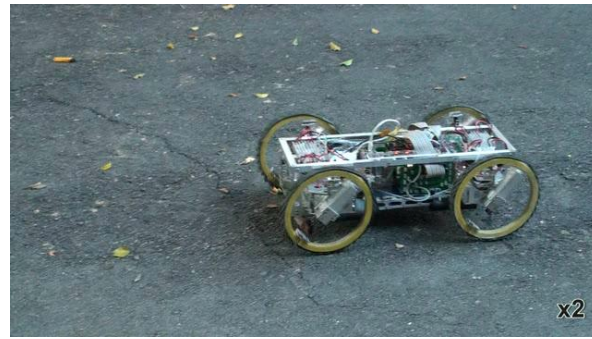
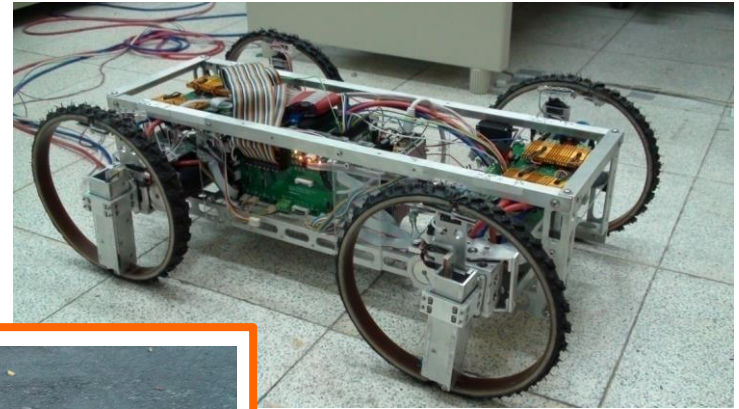
Actuator, Joint, and Cartesian Spaces -2

- Example: A leg-wheel transformable robot

輪模式

平地上

快速、平穩、省能



腳模式

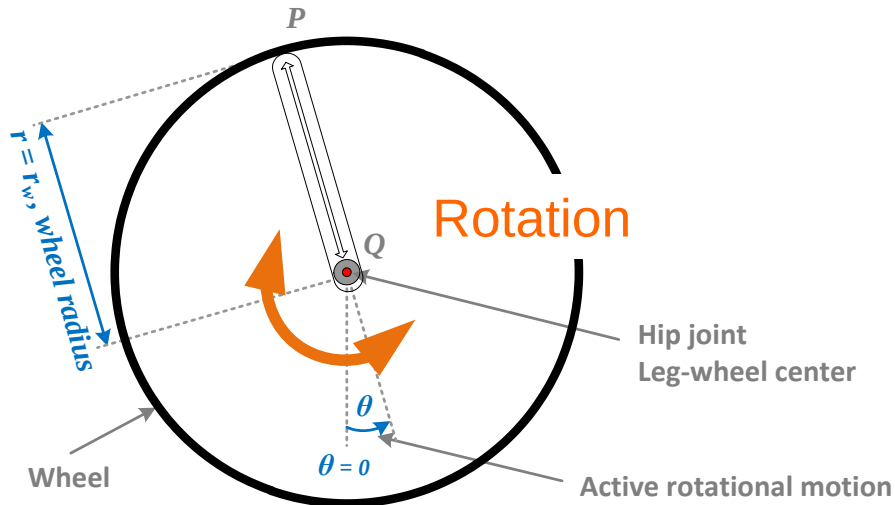
崎嶇地

越障、動態

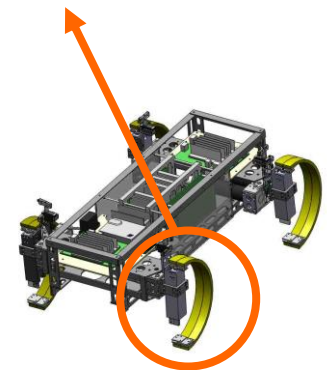
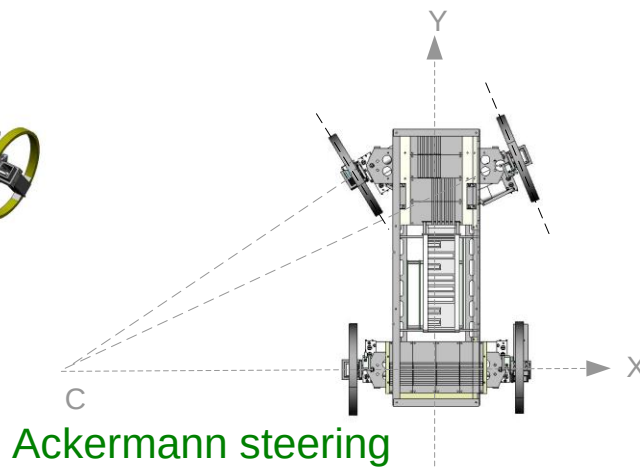
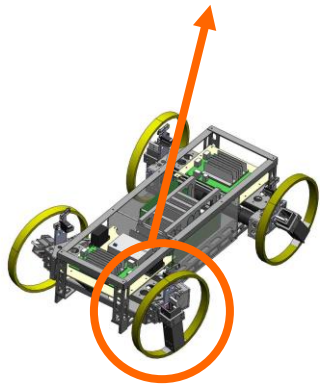
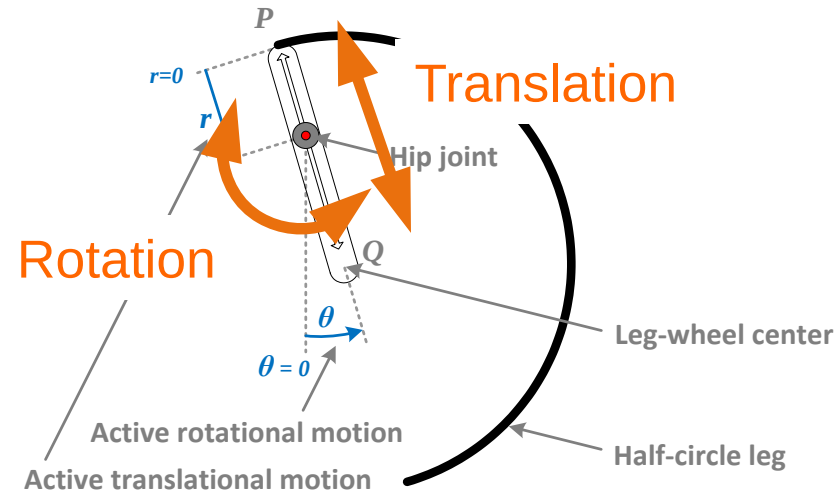
Actuator, Joint, and Cartesian Spaces -3



輪模式

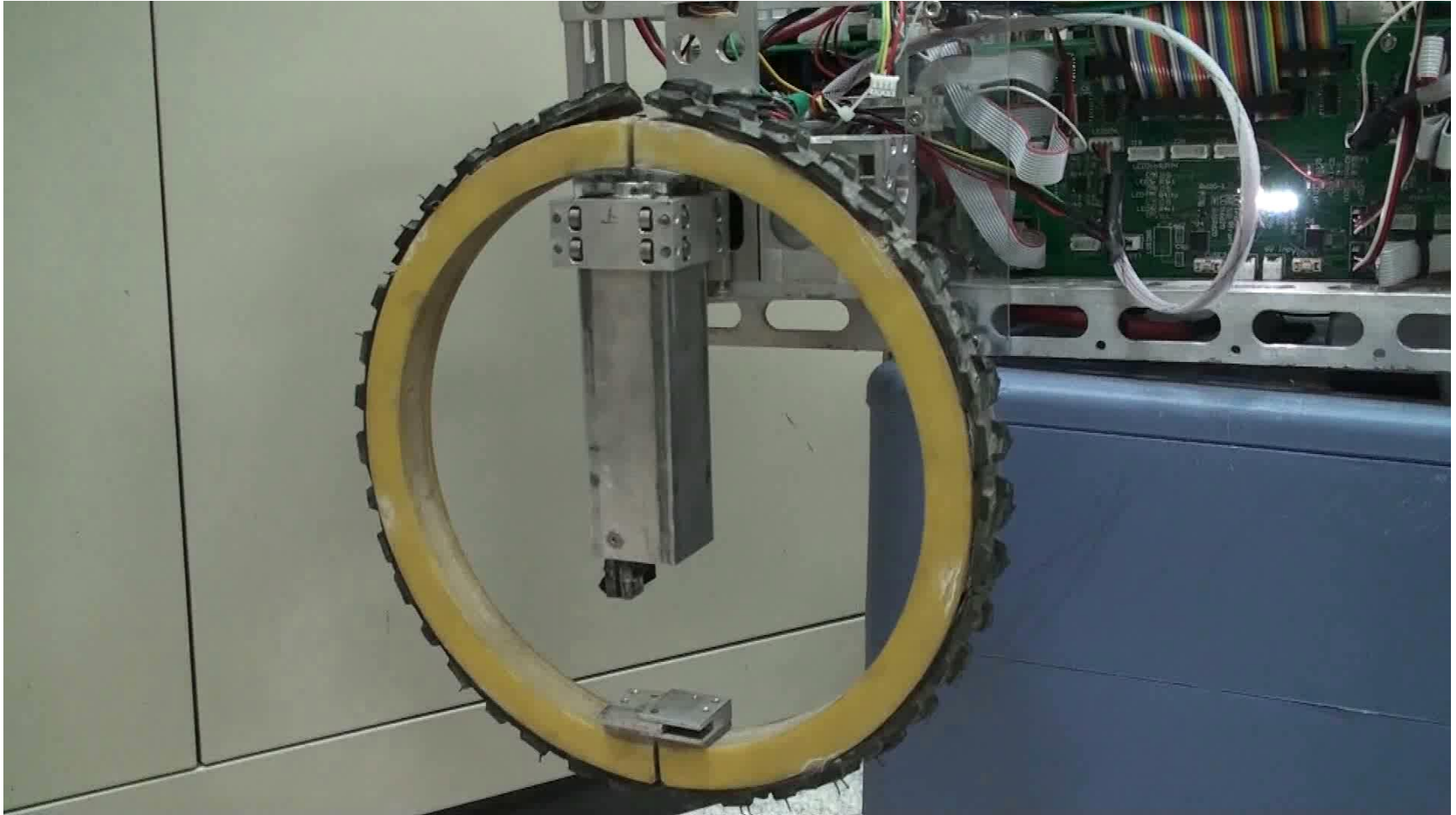


腳模式



Actuator, Joint, and Cartesian Spaces -4

- Leg-wheel motion



Actuator, Joint, and Cartesian Spaces -5

□ Kinematic mapping

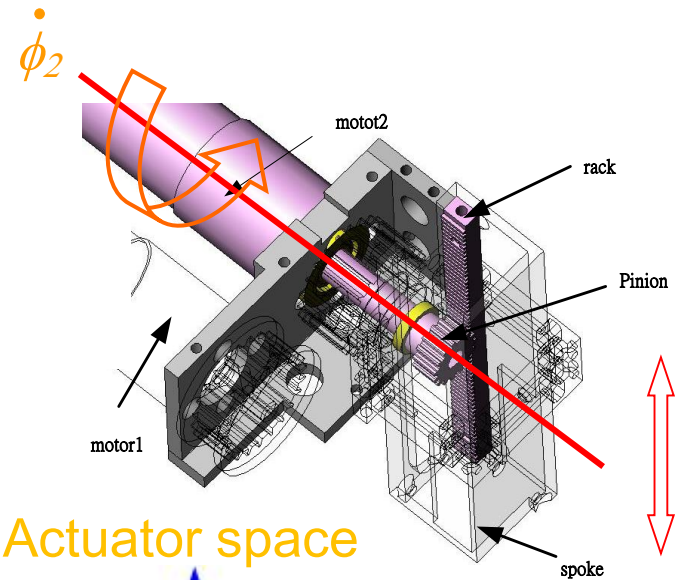
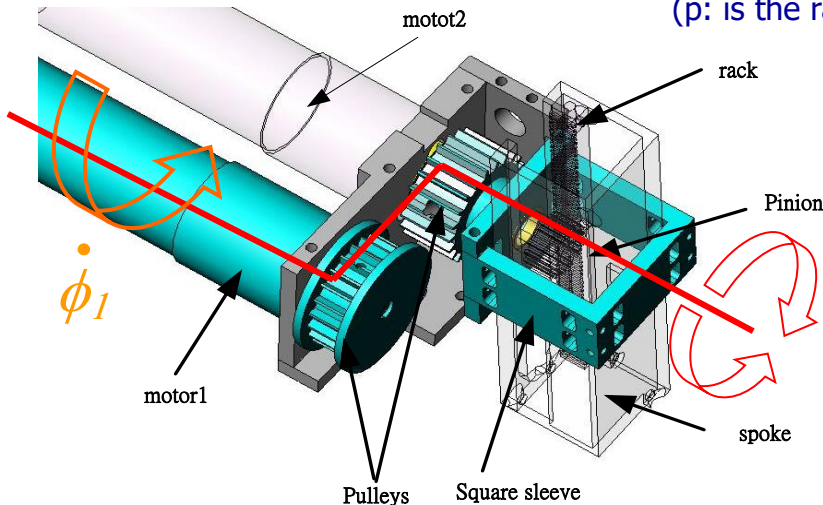
- ◆ Input: Motor speeds $\dot{\phi}_1$ $\dot{\phi}_2$
- ◆ Output: Leg-wheel motion $\dot{\theta}$ $\dot{r}$

in polar coordinate

Joint space $\dot{\xi} = \begin{bmatrix} \dot{\theta} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -p & p \end{bmatrix} \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix}$

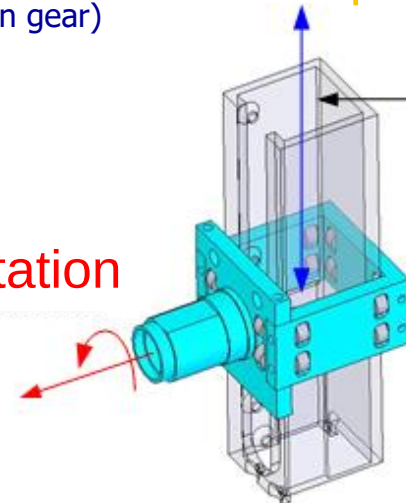
(p: is the radius of pinion gear)

Actuator space



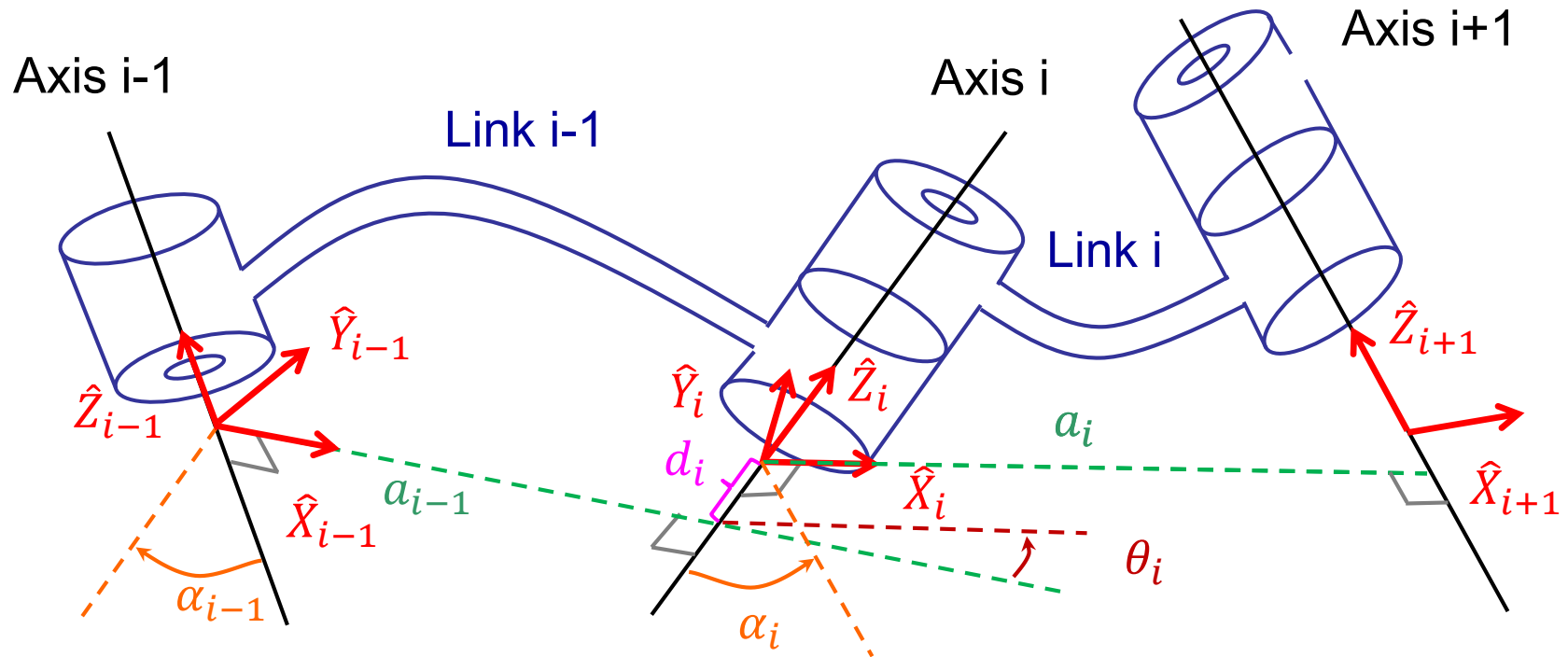
$\dot{\theta}$ Rotation

Translation $\dot{r}$



Denavit-Hartenberg 表達法小結 (Craig version) -1

- α_{i-1} : 以 $\hat{X}_{i-1}$ 方向看, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的夾角
- a_{i-1} : 沿著 $\hat{X}_{i-1}$ 方向, $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的距離 ($a_i > 0$)
- θ_i : 以 $\hat{Z}_i$ 方向看, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的夾角
- d_i : 沿著 $\hat{Z}_i$ 方向, $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的距離

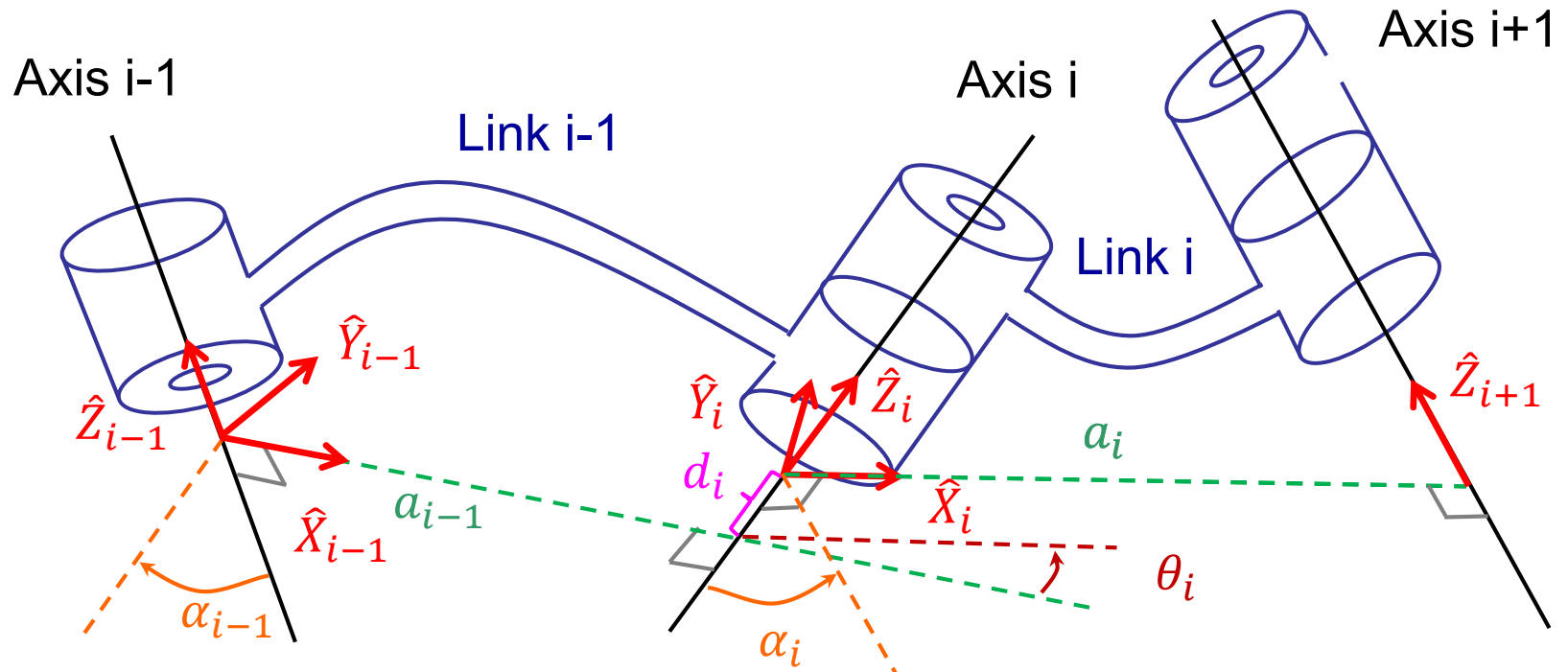


Denavit-Hartenberg 表達法小結 (Craig version) -2

$${}^{i-1}_iT = {}^{i-1}_R T {}^R_Q T {}^Q_P T {}^P_i T$$

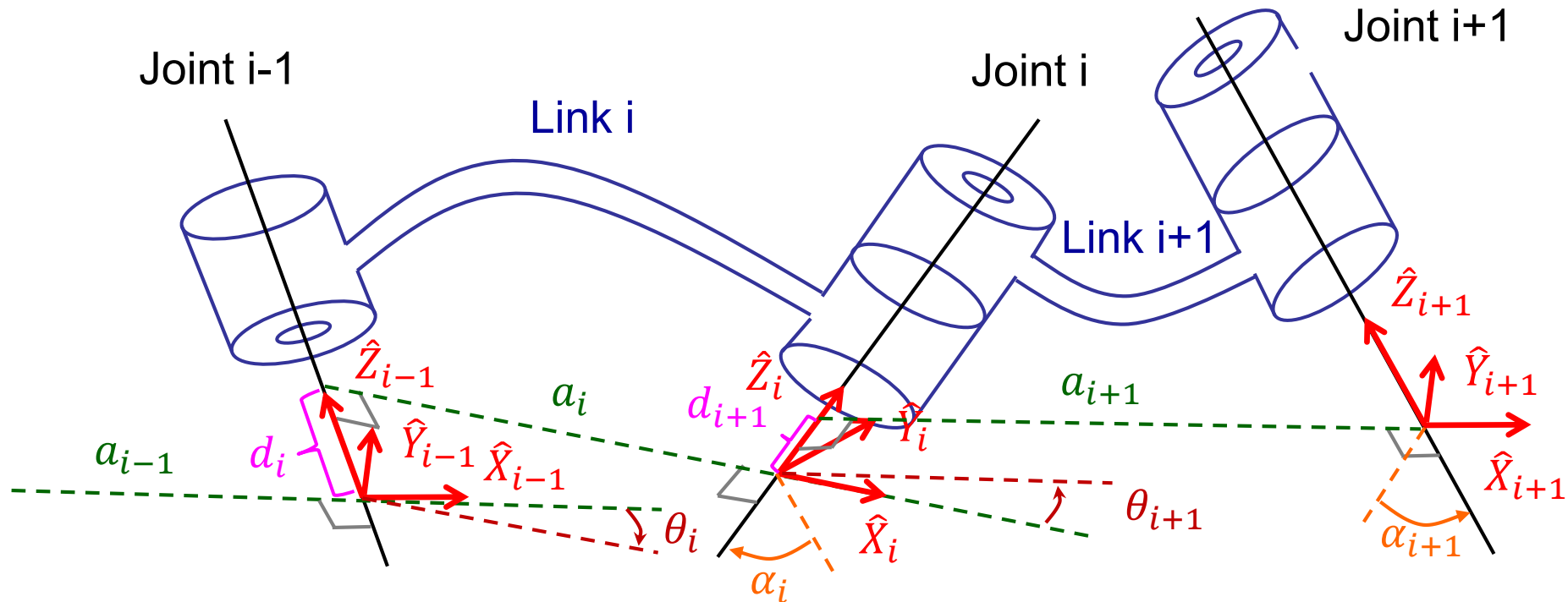
$$= T_{\hat{X}_{i-1}}(\alpha_{i-1}) T_{\hat{X}_R}(a_{i-1}) T_{\hat{Z}_Q}(\theta_i) T_{\hat{Z}_P}(d_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Denavit-Hartenberg表達法 (Standard) -1

- θ_i : 以 $\hat{Z}_{i-1}$ 方向看， $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的夾角
- d_i : 沿著 $\hat{Z}_{i-1}$ 方向， $\hat{X}_{i-1}$ 和 $\hat{X}_i$ 間的距離
- a_i : 沿著 $\hat{X}_i$ 方向， $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的距離 ($a_i > 0$)
- α_i : 以 $\hat{X}_i$ 方向看， $\hat{Z}_{i-1}$ 和 $\hat{Z}_i$ 間的夾角

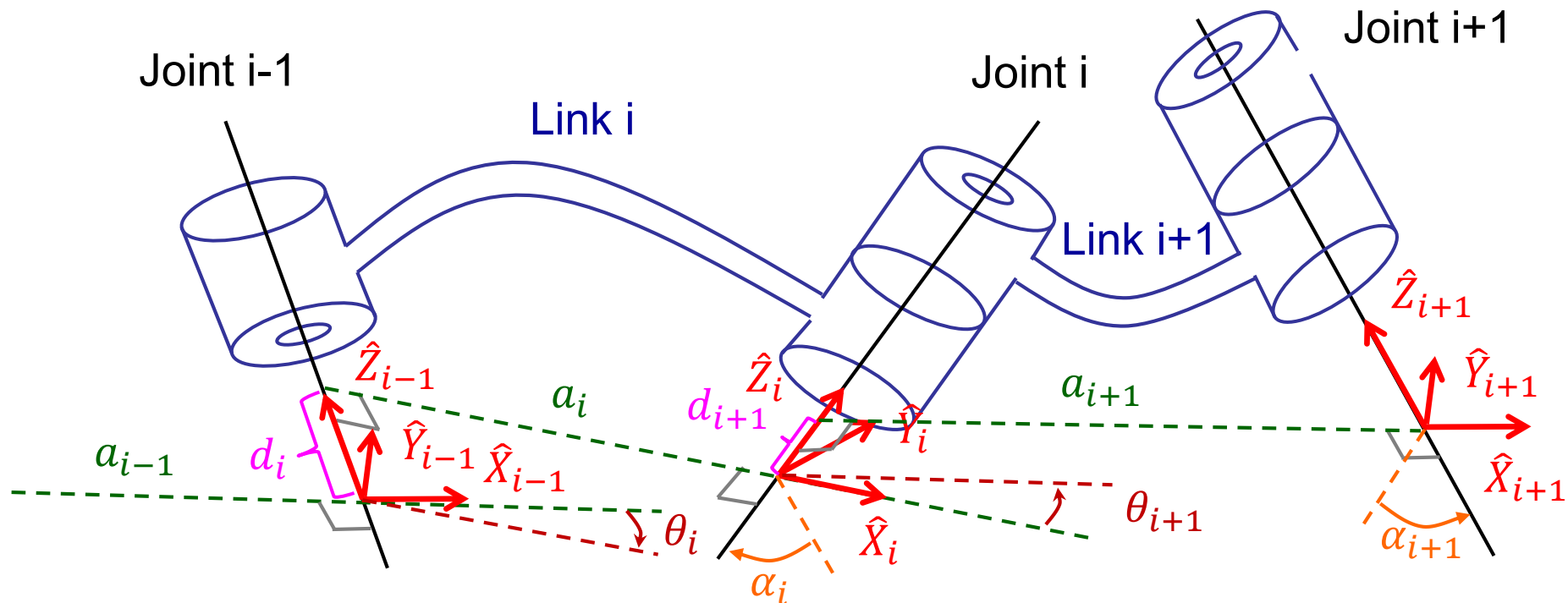


Summary of DH Notation (Standard) -2

$${}^{i-1}_{i}T = {}^{i-1}_R T {}^R_Q T {}^Q_P T {}^P_i T$$

$$= T_{\hat{Z}_{i-1}}(\theta_i) T_{\hat{Z}_R}(d_i) T_{\hat{X}_Q}(a_i) T_{\hat{X}_P}(\alpha_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Revisit Example: A RRR Manipulator -1

Craig DH

□ Joint axes

□ Common perpendiculars

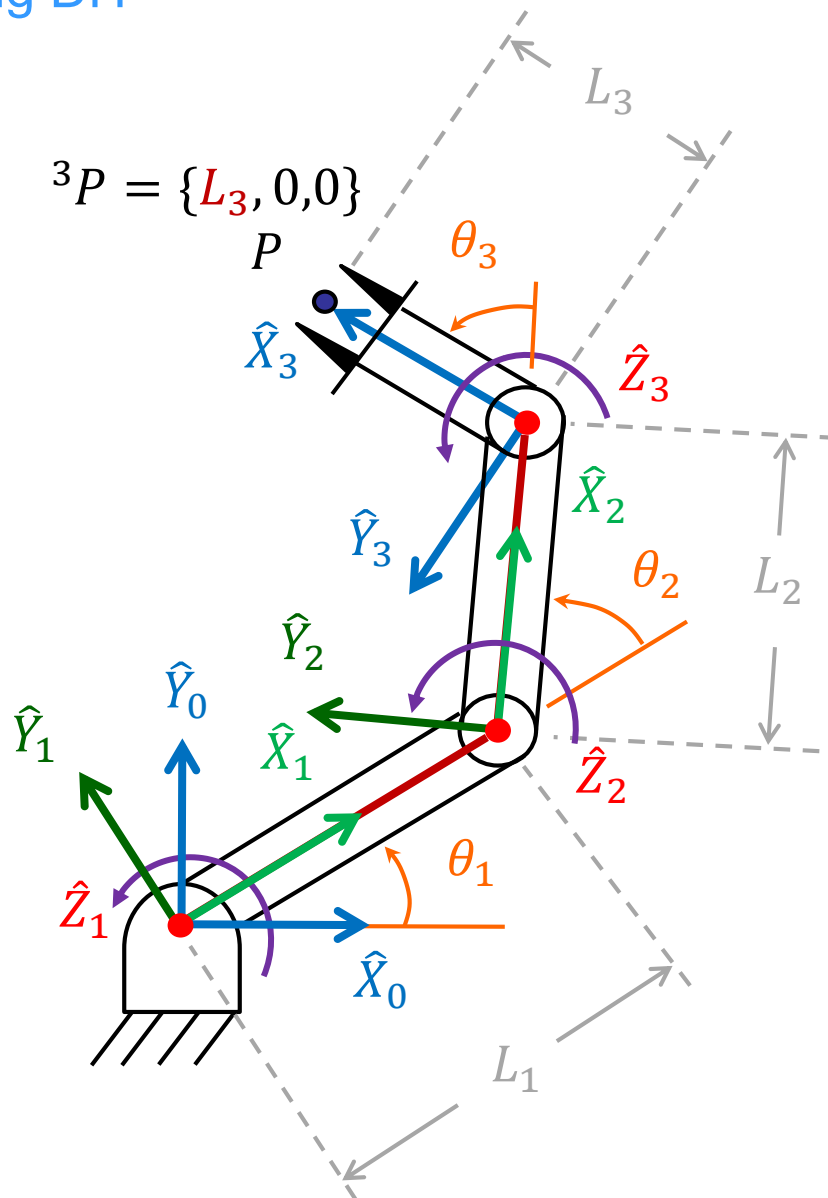
□ $\hat{Z}_i$

□ $\hat{X}_i$

□ $\hat{Y}_i$

□ Frames $\{0\}$ and $\{n\}$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



Revisit Example: A RRR Manipulator -2

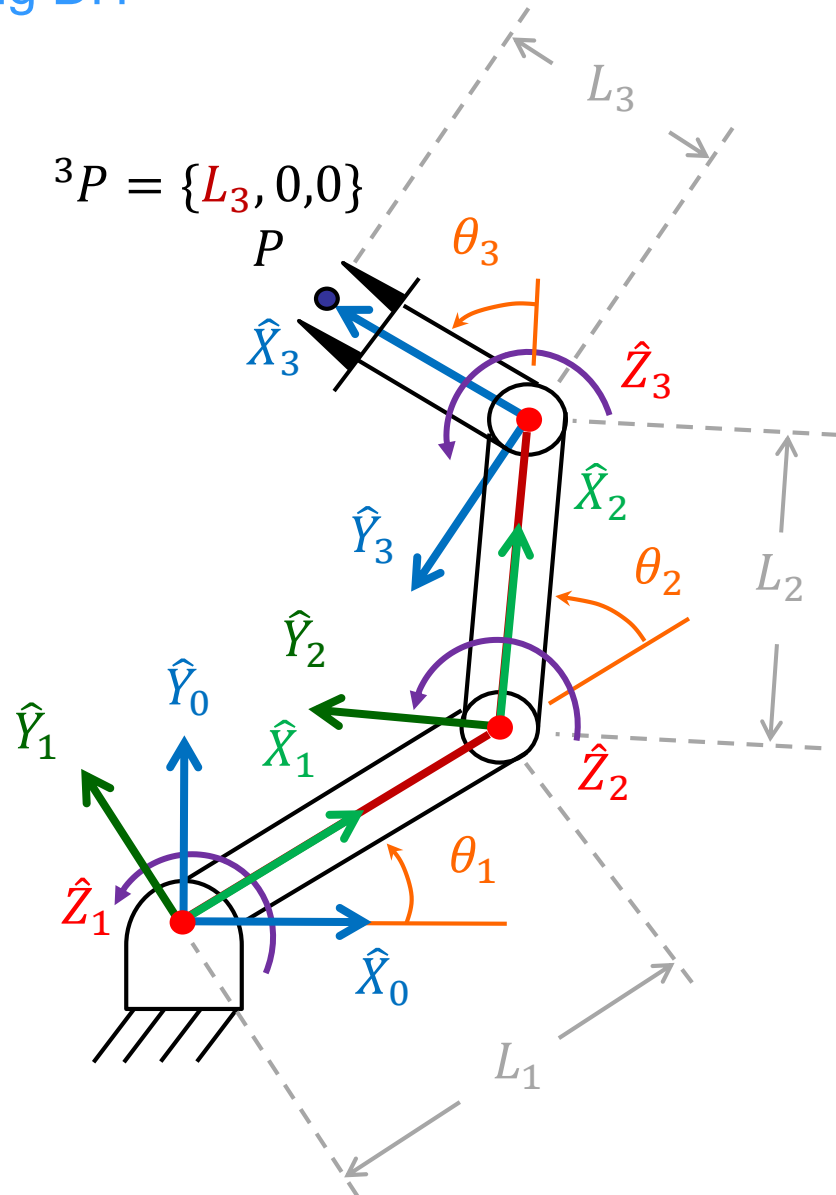
Craig DH

Transformation matrices

$${}^0_1T = \begin{pmatrix} \cos[t1] & -\sin[t1] & 0 & 0 \\ \sin[t1] & \cos[t1] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2T = \begin{pmatrix} \cos[t2] & -\sin[t2] & 0 & L1 \\ \sin[t2] & \cos[t2] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} \cos[t3] & -\sin[t3] & 0 & L2 \\ \sin[t3] & \cos[t3] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

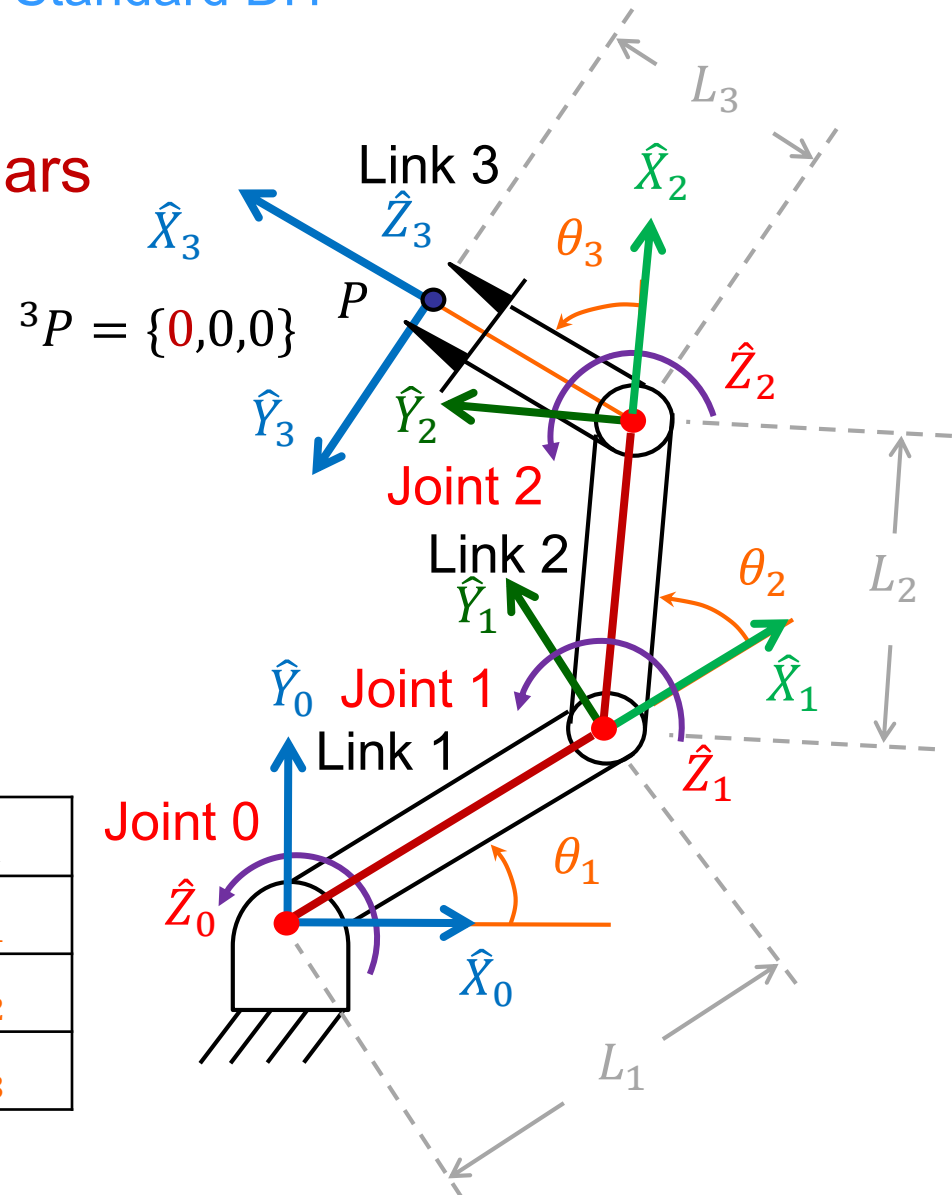


Revisit Example: A RRR Manipulator -3

Standard DH

- Joint axes
- Common perpendiculars
- $\hat{Z}_i$
- $\hat{X}_i$
- $\hat{Y}_i$
- Frames $\{0\}$ and $\{n\}$

i	α_i	a_i	d_i	θ_i
1	0	L_1	0	θ_1
2	0	L_2	0	θ_2
3	0	L_3	0	θ_3



Revisit Example: A RRR Manipulator -4

Standard DH

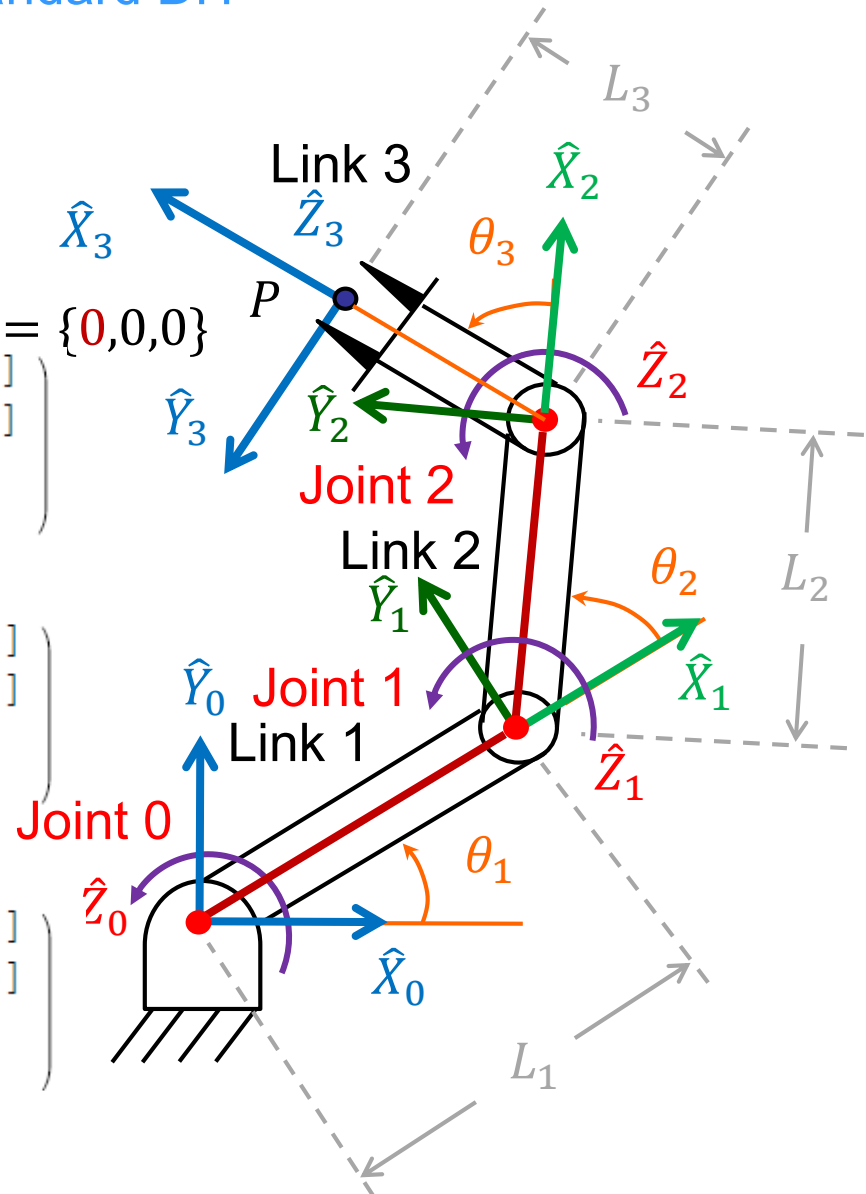
Transformation matrices

$${}^0_1T = \begin{pmatrix} \cos[t1] & -\sin[t1] & 0 & L1 \cos[t1] \\ \sin[t1] & \cos[t1] & 0 & L1 \sin[t1] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2T = \begin{pmatrix} \cos[t2] & -\sin[t2] & 0 & L2 \cos[t2] \\ \sin[t2] & \cos[t2] & 0 & L2 \sin[t2] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} \cos[t3] & -\sin[t3] & 0 & L3 \cos[t3] \\ \sin[t3] & \cos[t3] & 0 & L3 \sin[t3] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3P = \{0,0,0\}$$



Revisit Example: A RRR Manipulator -5

◆ Craig

$${}^0_3T \begin{pmatrix} \begin{matrix} \cos[t_1 + t_2 + t_3] & -\sin[t_1 + t_2 + t_3] & 0 \\ \sin[t_1 + t_2 + t_3] & \cos[t_1 + t_2 + t_3] & 0 \\ 0 & 0 & 1 \end{matrix} & \begin{matrix} L_1 \cos[t_1] + L_2 \cos[t_1 + t_2] \\ L_1 \sin[t_1] + L_2 \sin[t_1 + t_2] \\ 0 \end{matrix} \end{pmatrix}$$

$${}^0_3T \cdot T_{\hat{x}_3}([L_3, 0, 0])$$

$$\begin{pmatrix} \begin{matrix} \cos[t_1 + t_2 + t_3] & -\sin[t_1 + t_2 + t_3] & 0 \\ \sin[t_1 + t_2 + t_3] & \cos[t_1 + t_2 + t_3] & 0 \\ 0 & 0 & 1 \end{matrix} & \begin{matrix} L_1 \cos[t_1] + L_2 \cos[t_1 + t_2] + L_3 \cos[t_1 + t_2 + t_3] \\ L_1 \sin[t_1] + L_2 \sin[t_1 + t_2] + L_3 \sin[t_1 + t_2 + t_3] \\ 0 \end{matrix} \end{pmatrix}$$

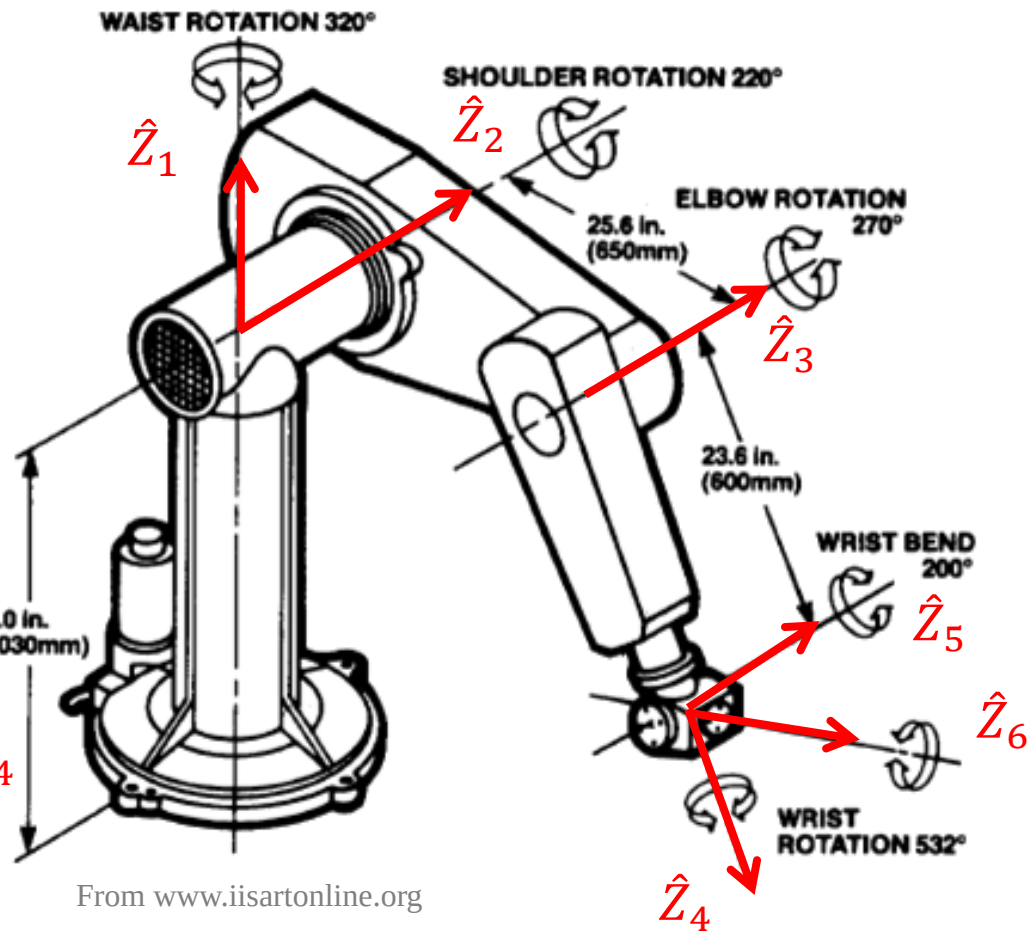
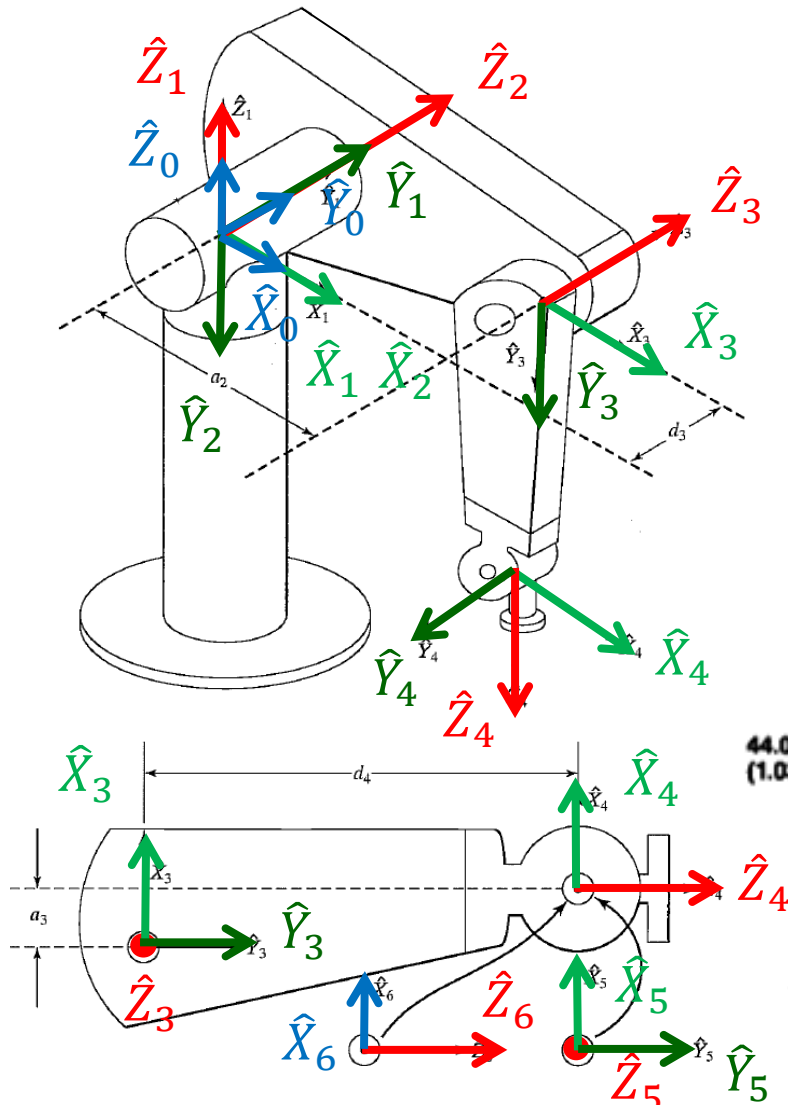
◆ Standard

0_3T

$$\begin{pmatrix} \begin{matrix} \cos[t_1 + t_2 + t_3] & -\sin[t_1 + t_2 + t_3] & 0 \\ \sin[t_1 + t_2 + t_3] & \cos[t_1 + t_2 + t_3] & 0 \\ 0 & 0 & 1 \end{matrix} & \begin{matrix} L_1 \cos[t_1] + L_2 \cos[t_1 + t_2] + L_3 \cos[t_1 + t_2 + t_3] \\ L_1 \sin[t_1] + L_2 \sin[t_1 + t_2] + L_3 \sin[t_1 + t_2 + t_3] \\ 0 \end{matrix} \end{pmatrix}$$

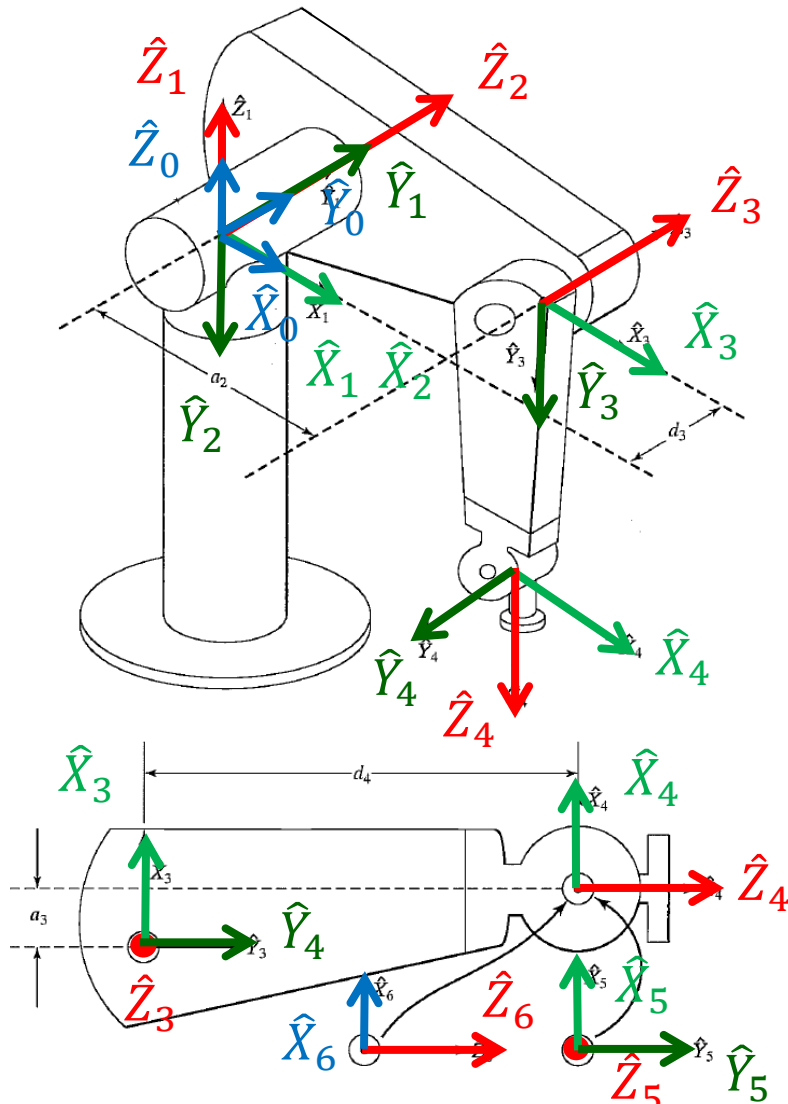
Example: PUMA 560 -1

□ Frames (Craig)



Example: PUMA 560 -2

□ DH parameters (Craig)



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	0	0	θ_2
3	0°	a_2	d_3	θ_3
4	-90°	a_3	d_4	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

Example: PUMA 560 -3

□ Transformation matrices

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s\theta_4 & -c\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

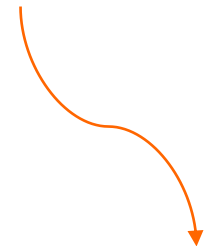
$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_6 & -c\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: PUMA 560 -4

□ Combining transformation matrices -1

$${}^4_6T = {}^4_5T {}^5_6T = \begin{bmatrix} c_5 c_6 & -c_5 s_6 & -s_5 & 0 \\ s_6 & c_6 & 0 & 0 \\ s_5 c_6 & -s_5 s_6 & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$


$${}^3_6T = {}^3_4T {}^4_6T = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & -c_4 s_5 & a_3 \\ s_5 c_6 & -s_5 s_6 & c_5 & d_4 \\ -s_4 c_5 c_6 - c_4 s_6 & s_4 c_5 s_6 - c_4 c_6 & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_3T = {}^1_2T {}^2_3T = \begin{bmatrix} c_{23} & -s_{23} & 0 & a_2 c_2 \\ 0 & 0 & 1 & d_3 \\ -s_{23} & -c_{23} & 0 & -a_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: PUMA 560 -5

□ Combining transformation matrices -2

$${}^1_6T = {}^1_3T {}^3_6T = \begin{bmatrix} {}^1r_{11} & {}^1r_{12} & {}^1r_{13} & {}^1p_x \\ {}^1r_{21} & {}^1r_{22} & {}^1r_{23} & {}^1p_y \\ {}^1r_{31} & {}^1r_{32} & {}^1r_{33} & {}^1p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1r_{11} = c_{23}[c_4c_5c_6 - s_4s_6] - s_{23}s_5s_6$$

$${}^1r_{21} = -s_4c_5c_6 - c_4s_6$$

$${}^1r_{31} = -s_{23}[c_4c_5c_6 - s_4s_6] - c_{23}s_5c_6$$

$${}^1r_{12} = -c_{23}[c_4c_5s_6 + s_4c_6] + s_{23}s_5s_6$$

$${}^1r_{22} = s_4c_5s_6 - c_4c_6$$

$${}^1r_{32} = s_{23}[c_4c_5s_6 + s_4c_6] + c_{23}s_5s_6$$

$${}^1r_{13} = -c_{23}c_4s_5 - s_{23}c_5$$

$${}^1r_{23} = s_4s_5$$

$${}^1r_{33} = s_{23}c_4s_5 - c_{23}c_5$$

$${}^1p_x = a_2c_2 + a_3c_{23} - d_4s_{23}$$

$${}^1p_y = d_3$$

$${}^1p_z = -a_3s_{23} - a_2s_2 - d_4c_{23}$$

Example: PUMA 560 -6

□ Combining transformation matrices -3

$${}^0_6T = {}^0_1T {}^1_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$r_{11} = c_1[c_{23}(c_4c_5c_6 - s_4s_5) - s_{23}s_5c_5] + s_1(s_4c_5c_6 + c_4s_6)$$

$$r_{21} = s_1[c_{23}(c_4c_5c_6 - s_4s_6) - s_{23}s_5c_6] - c_1(s_4c_5c_6 + c_4s_6)$$

$$r_{31} = -s_{23}(c_4c_5c_6 - s_4s_6) - c_{23}s_5c_6$$

$$r_{12} = c_1[c_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] + s_1(c_4c_6 - s_4c_5s_6)$$

$$r_{22} = s_1[s_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] - c_1(c_4c_6 - s_4c_5s_6)$$

$$r_{32} = -s_{23}(-c_4c_5s_6 - s_4c_6) + c_{23}s_5s_6$$

$$r_{13} = -c_1(c_{23}c_4s_5 + s_{23}c_5) - s_1s_4s_5$$

$$r_{23} = -s_1(c_{23}c_4s_5 + s_{23}c_5) + c_1s_4s_5$$

$$r_{33} = s_{23}c_4s_5 - c_{23}c_5$$

$$p_x = c_1[a_2c_2 + a_3c_{23} - d_4s_{23}] - d_3s_1$$

$$p_y = s_1[a_2c_2 + a_3c_{23} - d_4s_{23}] + d_3c_1$$

$$p_z = -a_3s_{23} - a_2s_2 - d_4c_{23}$$